

**AVAILABILITY, PRICE AND AFFORDABILITY OF  
ANTICANCER DRUGS IN NAIROBI, KENYA.**

**MBAYA LEWIS MUTUGI**

**U51/35847/2019**

*A thesis submitted in partial fulfilment of the requirements for award of  
the Degree of Master of Pharmacy in Pharmaco-epidemiology and  
Pharmacovigilance, University of Nairobi*

**DEPARTMENT OF PHARMACOLOGY, CLINICAL PHARMACY  
AND PHARMACY PRACTICE**

**UNIVERSITY OF NAIROBI**

**APRIL, 2024**

## UNIVERSITY OF NAIROBI DECLARATION OF ORIGINALITY FORM

|                     |                                                                             |
|---------------------|-----------------------------------------------------------------------------|
| Name of the student | Mbaya Lewis Mutugi                                                          |
| Registration number | U51/35847/2019                                                              |
| Faculty             | Health Sciences                                                             |
| Department          | Pharmacology, Clinical Pharmacy and Pharmacy Practice                       |
| Course work         | Master of Pharmacy in Pharmacoepidemiology and Pharmacovigilance            |
| Title of work       | Availability, Price and Affordability of Anticancer Drugs in Nairobi, Kenya |

I, Mbaya Lewis Mutugi, declare that:

1. I understand what plagiarism is and I am aware of the University's policy in this regard.
2. I declare that this thesis is my original work and has not been submitted elsewhere for examinations, an award, degree, publication. Where other people's work or my own work has been used, this has been acknowledged and referenced in accordance with the University of Nairobi requirements.
3. I have not sought or used the services of any professional agencies to produce this work.
4. I have not allowed and shall not allow anyone to copy my work with the intention of passing it off as his/her own work.
5. I understand that any false claim in respect of this work shall result in disciplinary action in accordance with the university plagiarism policy

Signature :



Date: 24<sup>th</sup> April, 2024

**Mbaya Lewis Mutugi U51/35847/2019**

## APPROVAL BY SUPERVISORS

This is to certify that this thesis has been submitted with our approval as the University Supervisors.

**PROF. F. A. OKALEBO**

Department of Pharmacology, Clinical Pharmacy and Pharmacy Practice

University of Nairobi

Signature.......... Date: 06/05/2024.....

**PROF. A. N. GUANTAI**

Department of Pharmacology, Clinical Pharmacy and Pharmacy Practice

University of Nairobi

Signature.......... Date: 20-05-2024.....  
...

**PROF. SYLVIA A. OPANGA**

Department of Pharmacology, Clinical Pharmacy and Pharmacy Practice

University of Nairobi

Signature.......... Date: 23/5/2024.....

## **ACKNOWLEDGEMENTS**

I acknowledge the Almighty God for His guidance throughout the course. I appreciate the support of my supervisors Prof. Faith Okalebo, Prof. Anastasia Guantai and Prof. Sylvia Opanga for their insight, guidance and inspiration to press on. Special thanks to my classmates with whom we have journeyed together.

## **DEDICATION**

I would like to dedicate this study to my mother, Josephine Nkatha, for always being my rock and motivating me through my academic life.

# TABLE OF CONTENTS

|                                                             |     |
|-------------------------------------------------------------|-----|
| UNIVERSITY OF NAIROBI DECLARATION OF ORIGINALITY FORM ..... | ii  |
| APPROVAL BY SUPERVISORS .....                               | iii |
| ACKNOWLEDGEMENTS .....                                      | iv  |
| DEDICATION.....                                             | v   |
| TABLE OF CONTENTS .....                                     | vi  |
| LIST OF TABLES.....                                         | ix  |
| LIST OF FIGURES .....                                       | x   |
| LIST OF EQUATIONS.....                                      | xi  |
| LIST OF ABBREVIATIONS AND ACRONYMS .....                    | xii |
| OPERATIONAL DEFINITION OF TERMS .....                       | xiv |
| ABSTRACT .....                                              | xv  |
| 1 INTRODUCTION .....                                        | 1   |
| 1.1 Background .....                                        | 1   |
| 1.2 Problem Statement .....                                 | 3   |
| 1.3 Research Questions .....                                | 4   |
| 1.4 Objectives.....                                         | 4   |
| 1.5 Justification .....                                     | 5   |
| 2 LITERATURE REVIEW .....                                   | 7   |
| 2.1 Global expenditure on Medicines .....                   | 7   |
| 2.2 Economic Impact of Cancer.....                          | 8   |
| 2.3 Determinants of Prices for Anticancer Drugs.....        | 9   |
| 2.4 Surveys on Pricing for Pharmaceuticals .....            | 13  |
| 2.5 Conceptual Framework .....                              | 17  |
| 3 METHODOLOGY .....                                         | 20  |
| 3.1 Study Design.....                                       | 20  |
| 3.2 Study Site .....                                        | 20  |
| 3.3 Target Population.....                                  | 20  |

|       |                                                                                               |    |
|-------|-----------------------------------------------------------------------------------------------|----|
| 3.4   | Sampling Frame and Procedure .....                                                            | 21 |
| 3.5   | Study Population .....                                                                        | 21 |
| 3.6   | Criteria for Selection of Survey Medicines .....                                              | 21 |
| 3.7   | Data Collection Instrument .....                                                              | 21 |
| 3.8   | Data Collection Procedures.....                                                               | 22 |
| 3.8.1 | Ghost Shopping Procedures .....                                                               | 22 |
| 3.9   | Quality Assurance .....                                                                       | 23 |
| 3.10  | Data Management .....                                                                         | 23 |
| 3.11  | Data Analysis .....                                                                           | 24 |
| 3.12  | Ethical Considerations .....                                                                  | 25 |
| 4     | RESULTS .....                                                                                 | 26 |
| 4.1   | Summary of Surveyed Anticancer Care Providers .....                                           | 26 |
| 4.1.1 | Types of Providers Surveyed .....                                                             | 26 |
| 4.1.2 | Procurement methods for anticancer drugs.....                                                 | 28 |
| 4.2   | Characteristics of The Surveyed Drugs .....                                                   | 29 |
| 4.3   | Availability of Anticancer Drugs At The Surveyed Providers .....                              | 30 |
| 4.4   | Availability of Originator Brands and Lowest Priced Generics .....                            | 32 |
| 4.5   | Prices of Anticancer Drugs At The Surveyed Providers .....                                    | 32 |
| 4.5.1 | Price Variation Analysis of Molecular Targeted Agents, Enzymes and Other Antineoplastics..... | 33 |
| 4.5.2 | Price Variation Analysis for Alkylating Agents, Detoxifying Agents and Antimetabolites .....  | 34 |
| 4.5.3 | Price Variation Analysis for Plant Derived agents, Antitumor antibiotics and Retinoids .....  | 36 |
| 4.5.4 | Comparison of Prices of Selected Anticancer Agents at the Supply Agencies                     | 37 |
| 4.5.5 | Comparison of Prices of Selected Anticancer Agents at the Hospitals .....                     | 37 |
| 4.5.6 | Comparison of Prices of Selected Anticancer Agents Across the Suppliers                       | 39 |
| 4.5.7 | Price Margin Analysis.....                                                                    | 40 |

|       |                                                                                                                 |    |
|-------|-----------------------------------------------------------------------------------------------------------------|----|
| 4.6   | Comparison of Median Unit Prices Obtained From The Three Main Sources With International Reference Prices ..... | 46 |
| 4.7   | Affordability of Anticancer Drugs In Kenya .....                                                                | 48 |
| 4.7.1 | Affordability of Breast and Prostate Cancer Treatments .....                                                    | 49 |
| 4.8   | Linear regression analysis for identification of determinants of prices of anticancer drugs in Kenya .....      | 51 |
| 5     | DISCUSSION .....                                                                                                | 53 |
| 5.1   | Availability of Anticancer Drugs .....                                                                          | 53 |
| 5.2   | Prices of Anticancer Drugs .....                                                                                | 55 |
| 5.3   | Affordability of Anticancer Drugs .....                                                                         | 58 |
| 5.4   | Determinants of Anticancer Drug Prices .....                                                                    | 59 |
| 5.5   | Study Strengths and Limitations .....                                                                           | 61 |
| 6     | CONCLUSION AND RECOMMENDATIONS. ....                                                                            | 62 |
| 6.1   | Conclusion .....                                                                                                | 62 |
| 6.2   | Recommendations for Policy and Practice .....                                                                   | 62 |
| 6.3   | Recommendations for Future Research .....                                                                       | 62 |
|       | REFERENCES .....                                                                                                | 63 |
| 7     | APPENDICES. ....                                                                                                | 69 |
| 7.1   | APPENDIX A: DATA COLLECTION FORM. ....                                                                          | 69 |
| 7.2   | APPENDIX B. ETHICAL APPROVAL. ....                                                                              | 72 |
| 7.3   | APPENDIX C: APPROVAL FOR WAIVER OF CONSENT. ....                                                                | 73 |
| 7.4   | APPENDIX D: AVAILABILITY OF ANTICANCER DRUGS.....                                                               | 81 |



## LIST OF TABLES

|                                                                                                           |    |
|-----------------------------------------------------------------------------------------------------------|----|
| Table 2.1. Factors contributing to key determinants of Anticancer Drug Prices. ....                       | 19 |
| Table 4.1 Dosage forms, Anticancer Classes and Formulations .....                                         | 29 |
| Table 4.2 Availability at Agencies, Hospitals and Supply Agencies .....                                   | 31 |
| Table 4.3 Price Variation Analysis for Molecular Targeted Agents, Enzymes and Other Antineoplastics ..... | 33 |
| Table 4.4 Price Variation Analysis for Alkylating Agents, Detoxifying Agents and Antimetabolites.....     | 34 |
| Table 4.5 Price Analysis for Plant Derived agents, Antitumor antibiotics and Retinoids                    | 36 |
| Table 4.6 Price Analysis at the Supply Agencies.....                                                      | 37 |
| Table 4.7 List of drugs that are more expensive in KNH.....                                               | 38 |
| Table 4.8 List of drugs that are cheaper at KNH.....                                                      | 38 |
| Table 4.9 Variability of prices across hospitals.....                                                     | 39 |
| Table 4.10 Price Ratios from Suppliers.....                                                               | 40 |
| Table 4.11 Price Margin analysis of KEMSA and KNH .....                                                   | 41 |
| Table 4.12 Price Margin analysis of KNH and Lowest Prices from Suppliers.....                             | 42 |
| Table 4.13 Price Margin analysis of Private hospital 3 and KEMSA prices .....                             | 44 |
| Table 4.14 Price Margin analysis for Private Hospital 3 against lowest prices from the suppliers.....     | 45 |
| Table 4.15 Comparison of Median Unit Prices of Selected Anticancer Agents .....                           | 47 |
| Table 4.16 Affordability of Selected Anticancer Drugs.....                                                | 49 |
| Table 4.17 Cost of drugs used for one cycle treatment of breast and prostate cancers.....                 | 50 |
| Table 4.18 Linear regression analysis for Determinants of anticancer drug prices .....                    | 51 |
| Table 7.1 Availability of Molecular Targeted Agents, Enzymes and Other Antineoplastic Agents .....        | 81 |
| Table 7.2 Availability of Alkylating Agents, Antimetabolites and Detoxifying Agents ..                    | 82 |
| Table 7.3 Availability of Plant Derived Agents, Antitumor Antibiotics and Retinoids ...                   | 83 |
| Table 7.4 Availability of Originator Brands and Lowest Priced Generics .....                              | 84 |

## **LIST OF FIGURES**

|                                                                                                       |    |
|-------------------------------------------------------------------------------------------------------|----|
| Figure 2.1. Conceptual Map of Determinants of Anticancer Drug Prices. ....                            | 18 |
| Figure 4.1 Relationship between the different providers of anticancer drugs in Nairobi,<br>Kenya..... | 27 |
| Figure 4.2 Procurement Methods Used to Acquire Anticancer Drugs.....                                  | 28 |
| Figure 4.3 Bar graph showing Availability of Originator Brands and Lowest Priced<br>Generics .....    | 32 |

## **LIST OF EQUATIONS**

|                                                                      |    |
|----------------------------------------------------------------------|----|
| Equation 3.1 Calculation of price Ratios.....                        | 24 |
| Equation 3.2 Calculation of Price Variation.....                     | 24 |
| Equation 3.3 Calculation of Price Margin.....                        | 24 |
| Equation 3.4 Calculation of the Expense of Anticancer Medicine ..... | 25 |

## **LIST OF ABBREVIATIONS AND ACRONYMS**

|       |                                          |
|-------|------------------------------------------|
| CA \$ | Canadian Dollars                         |
| DDD   | Defined Daily Dose                       |
| EMA   | European Medicines Agency                |
| EML   | Essential Medicines List                 |
| EU    | European Union                           |
| GDP   | Gross Domestic Product                   |
| HAI   | Health Action International              |
| KEML  | Kenya Essential Medicines List           |
| KEMSA | Kenya Medical Supplies Agency            |
| KES   | Kenya Shilling                           |
| KNH   | Kenyatta National Hospital               |
| LMICs | Low-and-Middle-Income-Countries          |
| LPG   | Lowest Priced Generic                    |
| MEDS  | Mission for Essential Drugs and Supplies |
| MSH   | Management Sciences for Health           |
| MUP   | Median Utility Price                     |
| NCD   | Non-Communicable Disease                 |
| NGO   | Non-Governmental Organization            |
| NHIF  | National Hospital Insurance Fund         |

|       |                                            |
|-------|--------------------------------------------|
| OB    | Originator Brands                          |
| TRIPS | Trade Related Intellectual Property Rights |
| USA   | United States of America                   |
| US \$ | United States Dollars                      |
| USD   | United States Dollars                      |
| WHO   | World Health Organization                  |

## OPERATIONAL DEFINITION OF TERMS

**Affordability:** A treatment is considered affordable if the course treatment of a medicine is not more than the daily lowest government wage.

**Availability:** Presence of medicines that are listed in the National Essential Medicines List, in a particular facility (hospital, supplier, wholesale/retail pharmacy). In this study, the medicines are 50 anticancer drugs.

**Co-payment:** This is a fee that a health insurer requires a patient to pay upon incurring a medical expense.

**National Essential Medicines List:** A list of drugs that have been deemed to satisfy the priority health care needs of the population in a country.

**Out-of-pocket:** This is whereby individuals pay for medical expenses using their own finances and is common in the absence of insurance.

**Price:** For the context of this study, a drug price is the fee paid by a patient to obtain a drug and may differ from one acquisition point to the other.

**Reference Pricing:** This is the approach of understanding the appropriateness of prices of medicines based on selected benchmark prices.

# **ABSTRACT**

## **Background**

Cancer is the second leading cause of death globally. In Kenya, cancer is the third leading cause of mortality after infectious and cardiovascular diseases. Cancer treatment is unaffordable in many countries especially low and middle-income countries. Anticancer drugs as a therapeutic category account for the highest expenditure for pharmaceuticals and this contributes to the inaccessibility of treatment for many patients. In Kenya, a survey of anticancer drugs across public, private sectors as well as the supply chain has not yet been conducted to establish availability, price and affordability.

## **Objectives**

The objective of this study was to assess the availability, price and affordability of anticancer drugs in Nairobi, Kenya.

## **Methodology**

A cross-sectional study was employed, focusing on suppliers dealing in anticancer drugs and hospitals offering anticancer treatment within Nairobi. Fifty anticancer drugs were assessed for availability, price and affordability in the three major sectors. Data was collected using standardized forms designed to capture information on the facility and the selected drugs. Prices obtained were compared to international reference prices set by Management Sciences for Health. Affordability was determined using World Health Organization/Health Action International Methodology. Data analysis was conducted using Microsoft Excel and the STATA Application (StataCorp, USA) version 13.1 software.

## **Results**

The list of 50 anticancer drugs surveyed yielded 92 different anticancer products due to varying formulations for individual drugs. Of the 92 products, there were 66 (71.74%) injections with tablets and capsule 26 (28.26%), making up the rest.

Availability of the target anticancer drugs was highest at the hospitals (50.8%) followed by suppliers (37.6%) and then the supply agencies (29.4%).

Linear regression conducted using alkylating agents as the reference class revealed biologics (KES 4875.4 (1096.4, 8654.4)) as the most expensive, followed by antitumour antibiotics (KES 2079.2 (659.2, 4817.5)) and finally plant derived agents (KES 223.8 (147.6, 595.2)). Of the 40 drugs available in both private and public hospitals, half were cheaper in private hospitals.

According to the World Health Organization/Health Action International criteria for determining affordability, none of the target drugs had a price lower than Kenya's daily lowest government wage (KES 411). Bendamustine 25mg had the highest monthly expense at KES 1065900 while Methotrexate 2.5mg had the lowest at KES 420.

## **Conclusion**

Availability of anticancer drugs was highest in the hospitals, followed by suppliers and finally supply agencies. Supply agencies should improve their stocks of anticancer drugs, which will translate to increased availability to public, and mission hospitals that serve a greater population. The findings also point to unaffordability of anticancer drugs, which limits access to treatment for most patients. Policy should be enacted by the Kenyan government regulating pricing in the pharmaceutical sector so as to reduce the burden of purchasing the drugs, for the consumers.



# 1 INTRODUCTION

## 1.1 Background

Non-communicable diseases are responsible for a majority of deaths globally and cancer is one of the leading causes (1). There were approximately 18.1 million new cancer cases and 9.6 million deaths as a result of cancer in 2018, with Asia accounting for 57.3% followed by Europe (20.3%), Americas (14.4%) and Africa (7.3%) (1). According to the World Health Organization, cancer is the second leading cause of death globally (2). The numbers are projected to rise with an estimated 22 million cases and 13 million deaths by 2030 (3). The increase reflects projected aging and growth and is set to affect low and middle-income countries more (3). As it stands, WHO estimates that 70% of deaths from cancer occur in LMICs (2). The high mortality has been attributed to late diagnosis and lack of appropriate funding to cater for treatment while the increase in incidence is as a result of the changes in the prevalence and distribution of risk factors associated with cancer as well as adoption of a Western culture of diet, physical inactivity, delayed reproduction and cigarette smoking (3, 4).

In 2018, estimates showed that there were 752,000 new cancer cases and 506,000 cancer deaths in Sub-Saharan Africa (5). In Kenya, cancer is third as a cause of mortality after infectious and cardiovascular diseases according to WHO (6). In 2011, there was an estimated incidence of 37000 cases and 28500 deaths, which grew to 47000 and 32987 respectively, by 2018 (6).

Globally, lung cancer is the most commonly diagnosed cancer followed closely by breast, colorectal and prostate cancers (1). The five most common cancers in men were lung, prostate, colorectal, stomach and liver while in women they were breast, colorectal, lung, cervical and stomach cancers (1). In Europe and North America, prostate and breast cancers are the most common cancers for men and women respectively with lung and colon cancer sharing an almost equal incidence for both sexes (7).

The incidence of cancers associated with the Western culture such as breast, colorectal, lung and prostate, have increased in LMICs (3). However, infection based cancers still have a high prevalence in these countries (3). In Sub-Saharan Africa, the most commonly diagnosed cancers in men are prostate and liver cancer whereas for women, breast and cervical have an almost equal incidence in this region (5). In Kenya, cancer is observed to affect more women than men and in 2018, 60% of the incident and 57% of the mortality cases were female (8). In the same year, the leading cancers affecting women in Kenya were breast, cervical and esophageal while for men they were esophageal, prostate and colorectal; for children, the most common were retinoblastoma, nephroblastoma, Burkitt's lymphoma and acute lymphoblastic lymphoma (8).

Cancer is one of the most expensive diseases to treat and its treatment is unaffordable in many countries (9). A major reason for this is its innate nature, pathophysiology and diversity. There exist more than 200 different types of cancer but there is no single standard treatment for them meaning that the resources dedicated to research and development are monumental. In addition, a single type of cancer can be treated with all major modes of treatment applied for cancer; surgery, radiotherapy and chemotherapy, consequently driving the cost up.

Another reason that makes cancer treatment unaffordable is the cost of anticancer drugs. Some view the production of anticancer drugs as a monopoly (10). Multiple anticancer agents do not compete in terms of which one is the most effective for a particular cancer but rather, they are used sequentially; if one fails to control the disease, the next drug is applied, therefore driving the costs up (11). A remedy for this would be entry of generic drugs once the patent of the originator drugs lapses; however, it has been observed that prices for anticancer generics are still significantly high (11).

This high cost of treatment, however, does not necessarily translate to improved outcomes or an increase in survival rates and in some cases; there is an inverse relationship between expenditure and outcomes (9). A study conducted in Europe found

that 39 out of 68 cancer drug indications approved by the EMA, entered the market without evidence of improved survival or quality of life (12).

Consequently, there has been a clamor to rationalize the costs of cancer treatment through various approaches such as value- based pricing using quality-adjusted life-years (11). Improvement of national guidelines to reflect the cost-effectiveness of various cancer treatments has also been suggested as a possible solution that would help patients and physicians make the correct treatment decisions (11). Government interventions to control the steep prices is another option as witnessed in Europe where some countries usually negotiate individually with pharmaceutical companies, but this is seen to weaken the EU's leverage as a region (13).

## **1.2 Problem Statement**

Anticancer drugs as a therapeutic category account for the highest expenditure for pharmaceuticals and the market for this category was predicted to grow to US \$150 billion by 2020 (14, 15).

In Sub-Saharan Africa where majority of economies are low-middle income, cancer treatment in general is inaccessible for most due to poor investment by most of the countries (19). Anticancer prices are still high as observed in developed countries.

The high cost of anticancer drugs which is witnessed globally, is usually passed onto the patients through co-payments or out-of-pocket expenditures. In the US, out-of-pocket expenses of one drug amounted to US \$20,000-30,000 per year which is nearly half of the average annual household income (20). In Sub-Saharan Africa, the average out-of-pocket expenditure on health is 40-50% ranging from 20% in South Africa and Botswana to 96% in Nigeria (19).

In Kenya, chemotherapy is estimated to cost between US\$ 60-600 per treatment course (4). In a largely uninsured population, this means out-of-pocket payments are made and

consequently treatment is inaccessible for most. Even for patients insured by the government's main insurance scheme (NHIF), there is a limit depending on the therapy. For example, a first line chemotherapy session is capped at Kshs 25,000 yet a session can cost up to Kshs 150,000 (21). A growing trend that has been witnessed over the recent past is patients diagnosed with cancer seeking specialized treatment in other countries, especially India where it is said to be cheaper.

In Kenya, there is no reference pricing therefore suppliers and retailers may charge prices that are above international reference prices. The extent of this practice remains unknown and even for the same products, there may be variations in local prices. The determinants of this variation are also unknown. Some critically important drugs such as biologics may be unavailable. This study aimed to address this knowledge gap by comparing prices of anticancer drugs in Kenya with international reference prices and assessing availability of selected anticancer agents. It is our hope that this study will contribute to improving access to medicines by highlighting the available products and their different prices within in the market.

### **1.3 Research Questions**

The research questions for this study included:

1. What is the variation in the price of anticancer drugs?
2. Are anticancer drugs available in Nairobi, Kenya?
3. Are the available anticancer drugs affordable?
4. What are the determinants of pricing for anticancer drugs in Nairobi, Kenya?

### **1.4 Objectives**

Main Objective

The main objective of this study was to assess the availability, price and affordability of anticancer drugs in Nairobi, Kenya.

## Specific Objectives.

The specific objectives were to:

1. Establish the availability and prices of anticancer drugs in supply agencies, suppliers and hospital pharmacies and compare local and international prices of anticancer drugs.
2. Assess the affordability of selected anticancer regimens amongst Kenyan patients.
3. Identify differences in prices and determinants of the pricing for anticancer drugs.

## 1.5 Justification

Anticancer drugs are the most expensive of any pharmaceutical product category. This makes them unaffordable and inaccessible especially for patients in LMICs. This study was a fact-finding mission to establish the availability, prices and end user affordability of the various anticancer drugs found in both public and private sector in Kenya. Results obtained were used to determine where we lie as a country with regards to availability and affordability of anticancer drugs as compared to other countries. In addition, they can inform a change in the current procurement practices especially in the public sector and this would trickle down to the patients in terms of cost reduction.

Stakeholders such as the National Cancer Institute of Kenya can use this study to advise the Ministry of Health on establishment of a policy that can improve access to anticancer drugs. Cancer patient and community support groups that make up the Kenya Network of Cancer Organizations, will also benefit from this information and can use it to lobby for price controls that would enable patients to access treatment.

Efforts will be made to publish the findings in a journal and this will be disseminated to the Ministry of Health, hospitals, patients and relevant stakeholders in the fight against cancer.



## **2 LITERATURE REVIEW**

### **2.1 Global expenditure on Medicines**

A significant proportion of every country's budget is allocated to funding health care. According to WHO, global spending on health was US \$7.8 trillion in 2017, which is about 10% of the GDP (22). It is believed that a significant proportion of this expenditure is spent on procurement of drugs; however, this depends on where a country ranks on the income scale. WHO estimates that high, upper-middle, lower-middle and low income countries spend 19.7, 23.1, 27.6 and 30.4% respectively, of their total health expenditure on pharmaceuticals (23).

In 2016, the US spent 17.9% of their GDP on health care and 10% of this was spent on prescription drugs (24). In 2010, the total health expenditure in China was 5% of GDP, with pharmaceuticals accounting for about 40% of this (25).

In high income countries, expenditure on medicines is mostly funded by public health insurance and social security systems with government spending accounting for 61.3% (23). However, in LMICs, 75% of expenditure is out-of-pocket which means that a significant part of the population is not able to access medicines (23). According to WHO, up to 90% of the population in developing countries obtain medicines through out-of-pocket payments, making medicines the second highest expenditure after food (26). The main determinants of pharmaceutical expenditure are medicine pricing and quantities consumed; therefore, appropriate policies regulating these two should be established especially in the case of middle and low income countries (23).

Most African countries are middle to low-income economies. The Abuja Declaration of 2001 had set a target for member states of the African Union to allocate at least 15% of their budgets to health care but for most, this has not been the case (27). Kenya is one of these states and in 2015/2016 financial year, government expenditure on health stood at

6.7% of total government expenditure (28). In the same financial year, out-of-pocket spending accounted for 27.7% of total health expenditure (28). This expenditure impedes health-seeking behavior of many Kenyans due to income limitations.

## **2.2 Economic Impact of Cancer**

The economic burden of cancer encompasses direct expenditure on health care and productivity losses due to morbidity and premature mortality (3). There are three cost categories: direct costs, productivity costs and intangible costs (3).

Direct costs arise from various resources that are applied in the detection, diagnosis and therapy of disease as well as post-treatment care such as rehabilitation and end-of-life care while productivity costs can be attributed to loss of economic resources and opportunities associated with cancer. Finally, intangible costs are those that cannot really be quantified but impact the quality of life of the patient, family and friends (3).

A report by the Economist estimated that the total economic cost of new cancer cases globally, was US \$286 billion in 2009 (29). Medical costs accounted for 53% of this figure whereas indirect costs accounted for 24% (29). In terms of anatomical sites, five cancers that accounted for 55% of the total costs are as follows: lung (US \$53 billion), colorectal (US \$33 billion), breast (US \$24 billion), prostate (US \$24 billion), and stomach cancers (US \$20 billion) (29).

According to a report from the American Cancer Society, in 2008, the total economic impact as result of cancer causing death and disability worldwide was US \$895 billion (30). This implied that cancer caused the highest economic loss as compared to other diseases (30). Morbidity and mortality from lung (US \$188 billion), colon (US \$99 billion) and breast cancer (US \$88 billion) accounted for a significant proportion of the economic costs. Cancers of the mouth, throat and cervix had the greatest economic impact (30).



In 2017, the US spent an estimated \$161.2 billion on cancer treatment, which resulted in a productivity loss of \$23.2 billion and a cost of premature mortality of \$150.7 billion (5). The aggregate figure is estimated to be 1.8% of the US GDP (5). In the EU, healthcare expenditure was €57.3 billion, with productivity losses due to cancer and premature death accounting for €10.6 billion and €47.9 billion, respectively (5).

In Canada, the economic burden of cancer care rose from CA \$2.9 billion in 2005 to CA \$7.5 billion in 2012 with hospital care cost making up a significant portion, followed closely by physician care and drug expenditure (31). Another study conducted in Brazil, estimated the total economic cost of cancer to be US \$ 49.5 billion in 2010 and US \$ 59.7 billion in 2015 (32). Of this total, direct costs accounted for 20%, morbidity costs 17% and mortality costs 63% (32).

In Kenya, the treatment cost for cancer is estimated to average between US \$ 1600 – 5000 (4). In a study conducted at the country's largest referral hospital, it was established that the average cost of treatment for cancer was US \$ 1413 (4). The greatest contributors to this high cost were hospitalization and anticancer drug charges (4).

### **2.3 Determinants of Prices for Anticancer Drugs**

Global expenditure on anticancer drugs is increasing at a very high rate and this is of concern to consumers, health professionals and governments (15). The global spending on cancer drugs and supportive care reached US \$133 billion in 2017, up from US \$90.9 billion in 2012, pointing to a 7.9% annual growth rate for these agents (33). It was estimated to grow and exceed US \$158 billion by 2020 (34). In the USA, the expenditure is estimated to grow at 10.3% annually (33). Some have suggested that the growth in expenditure is due to increase in the use of anticancer drugs, however, it is likely due to an increase in the prices of these agents (33).

Determinants of prices for anticancer drugs can be divided into 5 major categories. The first category includes costs incurred during the development and manufacture from an

industrial perspective, they include: (a) costs of research and development; (b) costs arising from production and commercialization of the product; (c) value of the drug to the end user and society; and (d) a return on investment to incentivize future research and development programs (33).

The pharmaceutical industry argues that the price of a drug has to cover expenses that went into the research and development of not only the approved drug but also other investigational candidates that might not have gotten approval (33). In addition, they also factor in the opportunity cost had that money been invested elsewhere (33), (35). This points to a gross overestimation as was seen in a research that estimated the total cost of developing a new drug to be US \$2.6 billion (14).

The cost of production includes expenses incurred during manufacturing, commercialization of the drug, approvals by regulators, distribution, sales and administration (33). Generally, the total cost of production takes up 50 - 70% of a pharmaceutical company's total income, with sales, marketing and administration costs accounting for around 30% of this (33). A significant amount of this expenditure is attributed to marketing and promotional activities, which if reduced could possibly result in lower operational costs and consequently a drop in drug prices (33).

The pharmaceutical industry tries to justify their prices on the basis of the return on investment needing to be enough to cater for research and discovery of future drugs (33). It has however, been observed that the companies usually generate enough revenue to cover their research and development costs in a few years. For instance, one study found that the total revenues generated by ten anticancer drugs since market authorization over a four year period was 9.3 times the total research and development expenditure (33).

Despite the factors discussed above, it was noted that a major determinant of the price is usually a company's expectation of the income that a drug can generate in the market (33). This has been supported by companies increasing prices of various drugs just to hit certain profit margins that have been set (33). An example is the increase in prices of two

anticancer molecules namely, bevacizumab and trastuzumab, by 5% and 8% respectively over two years with no change in production cost or value (33).

Secondly, the properties of the drug itself determine the price. These properties include whether the molecule is easy to synthesize, the type of formulation and whether the drug is a biological product. Intravenous formulations are generally more costly compared to oral formulations (15). In addition, biologics tend to cost more than chemical molecules. Molecules that have been in the market for long are usually cheap because of availability of generic products and in the case of biological products, presence of biosimilars provide cheaper alternatives (14). Taxane derivatives usually tend to be extremely costly as they are available in nature in small quantities thus cannot be completely synthesized (14). Semisynthetic processes are used on precursor molecules to manufacture different derivatives; however, the cost of production is still high (36).

A country's policies on drug pricing and procurement also play a key role as determinants (33). Regulatory agencies and purchasers of medicines such as national insurance schemes usually make decisions regarding these factors. Taxes and tariffs imposed by a country on importation of drugs can determine how expensive a drug is going to be. In addition, availability of policies and institutions that restrict or control pricing of pharmaceuticals affects drug prices (33). Price controls is whereby the relevant authorities in a particular country assess the proposed price of a medicine against the potential health benefit it is said to bring and its safety profile (33). In addition, they take into consideration local factors such as patterns of disease in particular populations, impact on patients and their disease relative to other available treatments, the country's health and economic needs (33). This process is, however, difficult to establish in the case of anticancer drugs as they do not have well established evidence to justify their clinical and economic values and this can be attributed to lack of data to demonstrate benefits in survival and well-being (33). In studies conducted to establish whether novel anticancer therapies justify their prices, it was found that there was no link between drug

cost and clinical benefit further questioning this valuation that is done to establish the price (37).

In Europe, regulatory agencies usually set prices based on comparison with the international reference price and sometimes directly negotiate with manufacturers for lower prices (13). In countries such as India and China, local policies have promoted local manufacturing of pharmaceuticals thus increasing access and effectively lowering the prices. In the US, policies require that the regulators conduct pharmaco-economic based assessments that evaluate cost effectiveness of drugs and limit registration based on findings (14).

The local supply chain in a country also determines a drug's price (33). The presence of many suppliers for anticancer products translates to a higher level of competition, which would lead to lower prices in contrast with just a single supplier for a given product. Operational costs incurred by suppliers also affect drug prices. In countries where taxes, salaries are high and aspects such as corruption play a role in day-to-day activities, drug prices end up being high. The level a supplier occupies on the supply chain also determines the prices at which they will sell their drugs. For instance, there can be a primary importer for a particular brand of drug who then sells to other distributors; therefore, obtaining the drug from the primary importer is cheaper than the other distributors. In addition, deliberate price segmentation by suppliers is sometimes witnessed whereby they set different prices for different types of consumers based on the socio-economic status of segments of the market.

The final category of determinants is the end users, which include facilities and the patients. At the facility level, the procurement method used can determine the prices. Public hospitals tend to procure by open tender system, which reduces prices while in the private sector, direct procurement is practiced and this translates to higher prices (33). The type of facility whether government, for-profit hospital or non-governmental determines prices of anticancer drugs. For example, Kenyatta National Hospital, which is

a government hospital, tends to have subsidized prices for the patients as compared to facilities that usually target patients of a high socio-economic class and therefore will have high prices (38). Oncologists and practitioners' preference for brands and tendency to prefer some medicines can cause use of expensive medicines as compared to cheaper alternatives.

According to doctors in the US, there are also other factors that account for the high costs of anticancer such as registration procedures emanating from various regulatory bodies; these increase costs and affect the life of a patent. Secondly, acceptance by oncologists of the high drug prices and readiness to adopt newer, more expensive drugs, which have minor improvements, and consequently, increases costs. In addition, the reluctance of oncologist and patient advocacy groups to lobby for lower drug prices pushes up drug prices. Finally, inflation of drug prices by intermediaries along the supply chain such as distributors, wholesalers, pharmacies and hospitals also pushes up drug prices (11).

## **2.4 Surveys on Pricing for Pharmaceuticals**

Affordable prices are recognized by WHO as one of the key elements in ensuring access to medicines alongside appropriate selection and use, sustainable financing and reliable health and supply systems (26). However, implementing policies that ensure affordable prices is still a challenge in many countries as they lack technical expertise to analyze and interpret the data on existing prices and local policies that would enable them to respond appropriately in case of price variations (26). This is observed especially in LMICs where prices are of importance due to the high level of out-of-pocket payments within the population.

In 2001, WHO member states endorsed a resolution to develop a standardized methodology that would be used to determine drug prices and this resulted in the launch of WHO/HAI Project on Medicine Prices and Availability (39). The project focused on development of standardized methods for measuring medicine prices, availability,

affordability and price components. After initial testing, the methodology was launched in 2003 (39). As a result, this method has been applied in over 50 countries for medicine price surveys (26).

A price survey conducted in Shaanxi Province of China using WHO/HAI methodology found that the prices, availability and affordability of medicines in the province should be improved to make sure that even the poor have access to basic treatment (40). A similar study that covered 36 developing and middle income countries found that prices in both public and private sectors for originator and generic medicines were significantly higher than expected and recommended that policies such as promoting generic medicines and alternative financing mechanisms are needed to increase availability, reduce prices and improve affordability (39).

In Africa, the WHO/HAI methodology has been applied in over 25 countries to determine availability and prices of medicines (41). A number of studies modified the methodology to conduct surveys that were specific to certain therapeutic classes such as malaria, antibiotics, NCDs and medicines for children (41). These therapeutic categories correspond to the disease burden in most sub-Saharan African countries. Since cancer is increasingly becoming a major cause of morbidity and mortality in Africa, studies specific to anticancer medicines should be conducted as they have in other parts of the world.

A survey conducted on the prices of 31 anticancer drugs in 18 countries located in Europe, Australia and New Zealand found price variations, with the price difference between the highest and lowest-priced drug ranging from 28-388% (42). Goldstein compared the differences between 8 patented anticancer drugs in high, middle and lower income countries and found that; the median cancer medicine price was highest in the United States (US \$ 8694) and was lowest in India (US \$ 1515), using the foreign exchange rates method (42). In terms of affordability, medicines in Australia were most

affordable based on the domestic product per capita while in India they were the least affordable (42).

In 2004, the Ministry of Health in Kenya supported by WHO and HAI, conducted a national survey of drug prices in public, private and NGO sectors. The study found that out-of-pocket payments for medicines was not affordable for a majority of the population and that there was a significant price variation within the three sectors (43). Another survey was conducted in 2018 to compare the price and availability of locally manufactured and imported medicines in Kenya using WHO/HAI methodology. This study found that locally produced products were on average lower priced compared to imported products and that the median prices of both locally and imported medicines were below the international reference prices (44).

Based on the WHO/HAI project, a number of policies were identified as being the major determinants of affordability and availability in LMICs (26). Regulation of mark-ups to medicine prices in the pharmaceutical supply and distribution chain was identified as one of the policies. Mark-ups are additional costs applied to the price of the commodity to cover overheads, distribution expenses and profit. If not regulated, these usually lead to high medicine prices and affect affordability as costs are added on as they move from wholesale to retail and along the supply chain (26). Secondly, medicines should be exempted from certain taxes imposed by the government to ensure affordable prices. In addition, international reference pricing can be used to set the price of a product in a given country as this ensures some uniformity. Finally, countries are encouraged to promote use of quality assured generic medicines which are usually cheaper as compared to originator products (26).

The WHO/HAI studies have seen various governments take action to regulate medicine prices. In Indonesia, prices of 458 generic medicines were reduced by 5-70% and regulations were implemented to standardize purchasing. In Lebanon, medicine prices as

well as markups were significantly reduced and in Nigeria, they decided to draft policy based on surveys that have been conducted (45).

Studies that are specific to certain therapeutic categories have been conducted in Africa as mentioned earlier but not for anticancer drugs. The burden of cancer incidences in sub-Saharan Africa is expected to grow to over 85% by 2030 (46). In Kenya specifically, there is growing demand for cancer treatment but limited supply capacity with regards to treatment (46). At the time of conducting this study, to the best of our knowledge, no study had yet looked at the commodity aspect of cancer treatment. Most studies focused on treatment of specific types of cancers but not drugs recommended in the essential medicines list. Therefore, this study shall play a key role in shedding light on the drugs recommended to treat cancer with respect to their availability, price and affordability. This information shall set a basis for further research in the country and transparency of drug pricing.



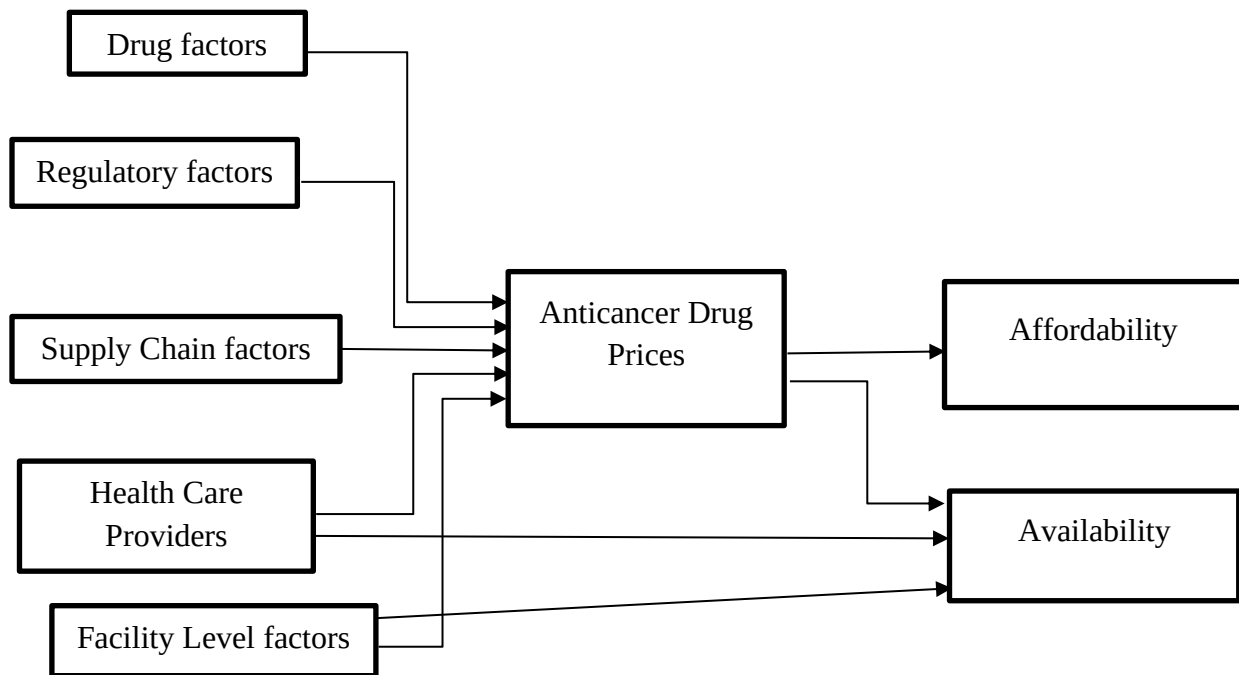
## 2.5 Conceptual Framework

The conceptual framework of this study was based on the key determinants of anticancer drug prices as illustrated in the conceptual map on *Figure 2.1*. Drug factors that contribute to the price of anticancer drugs include the type of formulation; the drug being a biologic, synthetic or from a natural source and the presence of biosimilars in the market, in the case of biologics. Regulatory factors include policies that promote local drug manufacturing, existence of laws that control drug prices and adherence to international trade agreements such as TRIPS. Supply chain factors influencing the prices include operational costs incurred by a company; the level at which the supplier sits on the hierarchy of distribution for the particular drugs. Health care providers also influence drug prices by having preferences of specific brands for treatment of the different cancers and by generally accepting the prevailing drug prices for anticancer drugs in the market. Facility level factors that affect drug prices include whether the hospital is private or public; the type of procurement method used by the facility for example, tender vs direct procurement and finally, the catchment area of the respective hospital. Hospitals located in urban areas where people have a high socio-economic standing will tend to price their drugs a bit higher compared to the rest. For this particular study, only three determinants will be measured, namely: drugs, supply chain and facilities.

The anticancer drug prices directly have an impact on the affordability of a particular treatment course. According to WHO/HAI methodology, a treatment is considered affordable if the cost of a course of treatment of a medicine is not more than the daily lowest government wage (45).

In addition, prices also affect the availability of drugs. Cancer care providers may opt not to stock expensive drugs due to projected losses if the drugs are not used. Generic drugs, which are cheaper than originator brands, may as a result be more available. A number of factors discussed above also play a role in affecting availability independent of price. For example, private facilities may procure more drugs due to their financial capabilities and

as a result may have more drugs available as compared to public facilities. Health care providers may have a preference of using certain drugs over others hence influence the overall availability. Table 2.1 outlines the factors that contribute to the different determinants discussed above.



*Figure 2.1. Conceptual Map of Determinants of Anticancer Drug Prices.*

Table 2.1. Factors contributing to key determinants of Anticancer Drug Prices.

| Determinants          | Factors Contributing to Cost                                                                                                                                                                                                         |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Drugs                 | <ul style="list-style-type: none"> <li>- Intravenous vs Oral Formulation</li> <li>- Biologic vs Chemical molecules</li> <li>- Availability of Biosimilars</li> <li>- Natural vs Synthetic molecules</li> </ul>                       |
| Regulatory            | <ul style="list-style-type: none"> <li>- Policy promoting local manufacture</li> <li>- Existence of drug price controls</li> <li>- Adherence to TRIPs agreement</li> <li>- Parallel Importation.</li> </ul>                          |
| Supply Chain          | <ul style="list-style-type: none"> <li>- Operational costs</li> <li>- Operating level of supplier (Importer, distributor, Retailer)</li> <li>- Level of competition (Number of suppliers)</li> <li>- Differential pricing</li> </ul> |
| Health Care Providers | <ul style="list-style-type: none"> <li>- Preference for specific brand</li> <li>- General acceptance of high prices</li> </ul>                                                                                                       |
| Facility              | <ul style="list-style-type: none"> <li>- Private vs Public hospital</li> <li>- Procurement method used</li> <li>- Catchment area of the Hospital</li> </ul>                                                                          |

### **3 METHODOLOGY**

#### **3.1 Study Design**

A cross-sectional study design was employed in January 2022, to obtain information from the different establishments handling anticancer drugs. This method was selected because it provided generation of real-time data on cancer care providers within Nairobi. It entailed a survey of medicine prices as described in WHO/HAI methodology.

#### **3.2 Study Site**

The study was based in Nairobi, Kenya because it houses most of the country's suppliers and the majority of hospitals that offer anticancer care. In addition, the two main supply agencies, KEMSA and MEDS, have their headquarters domiciled in Nairobi. According to one study, most facilities offering cancer care are concentrated within a 5km radius of each other in Nairobi therefore, limiting access to many patients seeking treatment since 78% of Kenyans are rural residents (46).

#### **3.3 Target Population**

The target population was supply agencies, suppliers dealing in anticancer drugs and hospitals offering anticancer treatment within Nairobi. Wholesalers and retailers were merged into the suppliers' category as it was observed that some establishments used this labels interchangeably depending on the buyer. The estimated numbers of suppliers and hospitals was 20 and 10 respectively. There are two main drug supply agencies in Kenya and both were included in the study. The public sector accounts for 20% whereas the private sector accounts for 80% (46).

### **3.4 Sampling Frame and Procedure**

Purposive sampling was done for supply agencies, suppliers and hospitals that deal with anticancer drugs. This method of sampling was selected because it was able to focus on the providers who stocked anticancer drugs. The process entailed looking at price lists for supply agencies and suppliers to determine whether they stocked anticancer drugs and calling different hospitals to determine whether they offer cancer treatment.

### **3.5 Study Population**

The purposive sampling resulted in a sample comprising two supply agencies, thirteen suppliers and ten hospitals. However, challenges such as non-response bias were encountered especially when it came to hospitals and as a result, responses were only obtained from four hospitals.

### **3.6 Criteria for Selection of Survey Medicines**

Kenya Essential Medicines List 2019 was used as the source document (47). This is because the guidelines provide a list of minimum medicine supplies that must be available at the specified facility level all the time. This resulted in selection of 46 drugs. In addition, four drugs that are not in the essential medicines list were included to establish their availability in the country as well. These four additional agents are molecular targeted agents, which are increasingly being applied in treatment of cancer. They are included in the Kenya Cancer Treatment Protocols 2019 but are not on the Essential Medicines List (48).

### **3.7 Data Collection Instrument**

A custom designed data collection form (Appendix A), was used. The information that was collected included: facility/institution information (name, location, type of facility);

medicine information (dosage formulation, strength, unit price, medicine type). The form also had a section that sought to establish the avenue by which drugs are procured by the respective facility.

### **3.8 Data Collection Procedures**

Two research assistants obtained prices from suppliers by requesting their sales departments for the price lists, which are usually provided to customers. Prices from hospitals were obtained by ghost shopping which involved using individuals who posed as patients who are interested in the drugs for treatment and in doing so, they obtained the prices.

#### ***3.8.1 Ghost Shopping Procedures***

Ghost shopping, also known as mystery shopping, is a research method commonly applied in marketing research to establish various aspects of service offered by a provider. It involves using trained shoppers who access the specified product or services and report back with detailed and objective information (49) . This method has also found application in health research. For instance, mystery shopping was applied to assess patient counselling for the common cold in community pharmacies in Slovakia (50).

For this particular study, this technique was employed to establish drug prices from hospitals. The reason for opting for this method is because; most hospitals were not willing to reveal prices of their drugs due to competition and other reasons. Five final year undergraduate pharmacy students were recruited as the ghost shoppers because of their knowledge of anticancer drugs. The list of drugs provided was divided into different treatment regimens for different types of cancers as would appear on a prescription. The students were then trained by the principal investigator to inquire for the drug prices by simulating different scenarios and role-playing amongst themselves. In

order to establish their readiness, a test run was conducted by the principal investigator and after confirming their readiness, they were deployed to the field to commence the study.

### **3.9 Quality Assurance**

Research assistants were recruited to help with data collection due to the number of sites involved in the study. The assistants were final year pharmacy students who had knowledge of anticancer drugs. Their main role was data collection whereas the principal investigator conducted their training and coordination during the data collection process. The assistants underwent training on what the study sought to establish and on the different anticancer drugs involved. Thereafter, they were trained on good data collection practices and on data collection using the form in Appendix A. The form was pretested at the Kenyatta National Referral Hospital. During the study, the research assistants checked for the completeness and consistency of the information collected at the end of each day, using filled pretest forms as references.

### **3.10 Data Management**

At the end of each week, the data collected by the research assistants was forwarded to the principal investigator who then entered it into Microsoft Excel 2016 Workbooks and saved it on a password-protected laptop. This served as an extra step in verifying the data collected by the assistants. A copy of the data was made and stored in a flash drive that was kept under lock and key for backup and security purposes.

Three Microsoft Excel Workbooks were created to store data for suppliers, agencies and hospitals. These workbooks were stored in a common folder. The data entered was cleaned and coded prior to analysis being done.

### 3.11 Data Analysis

Summary data analysis was conducted. The drugs surveyed were analyzed according to dosage forms, anticancer class and formulations.

The availability of all anticancer medicines surveyed was determined for each of the surveyed providers. In addition, availability of each drug surveyed was also determined across all providers.

Prices of the different anticancer drugs were obtained during the survey. Maximum and minimum prices were obtained for all surveyed drugs, from all providers surveyed and price variation analysis was conducted using the formulas presented in equations 3.1 and 3.2. Price margin analysis was also conducted to determine the various markups added onto drugs before reaching the patients using equation 3.3.

$$\text{Price ratio} = \text{Maximum price} / \text{Minimum price}$$

*Equation 3.1 Calculation of price Ratios*

$$\text{Percentage price variation was calculated by, Price Variation (\%)} = [\text{Maximum Price} - \text{Minimum Price} / \text{Minimum Price}] \times 100$$

*Equation 3.2 Calculation of Price Variation*

$$\text{Percentage Price Margin (\%)} = [\text{Hospital Price} - \text{Supplier Price} / \text{Supplier Price}] \times 100$$

*Equation 3.3 Calculation of Price Margin*

Linear regression was conducted to assess the determinants of anticancer drug prices. This analysis examined the impact of pharmacological classes, dosage forms, facility type and originator brands on the drug prices.

As mentioned earlier, as per WHO/HAI methodology, a treatment is considered affordable if the course treatment of a medicine is not more than the daily lowest



government wage. The affordability per month was determined by the formula presented in equation 3.3 (51):

$$\text{Expense of anticancer medicine} = \text{Median Unit Price (MUP)} \times \text{Defined Daily Dose (DDD)} \times 30 \text{ Days.}$$

*Equation 3.4 Calculation of the Expense of Anticancer Medicine*

The WHO/HAI methodology recommends that for acute conditions, the duration of the full course of therapy be applied while for chronic conditions where treatment might continue indefinitely, a one-month course of therapy be applied (45). Cancer falls in the latter category, hence the application of 30 days to show the economic implication on patients even though they are expected to have more than cycle of treatment with multiple medicine regimens.

To determine affordability, the lowest priced option out of the three providers was used to calculate the monthly expense of each drug. In addition, the median unit prices for the drugs were compared to international reference prices for each drug using the International Medical Products Price Guide 2015 provided by MSH (52).

### **3.12 Ethical Considerations**

Ethical approval to conduct this study was sought and obtained from the Kenyatta National Hospital-University of Nairobi Ethics and Research Committee. Letter for ethical approval is attached in Appendix B, Ref: KNH-ERC/A/487.

Due to the nature of the study, a waiver of informed consent was sought, obtained and is attached in Appendix C. This was to enable the researcher avoid falsified responses from the different facilities and institutions who were not forthcoming with the information if made aware of the study. Once data was collected, identities of the different institutions was not revealed and instead references such as “Private Facility 1” or “Supplier 1” was used.

## 4 RESULTS

### 4.1 Summary of Surveyed Anticancer Care Providers

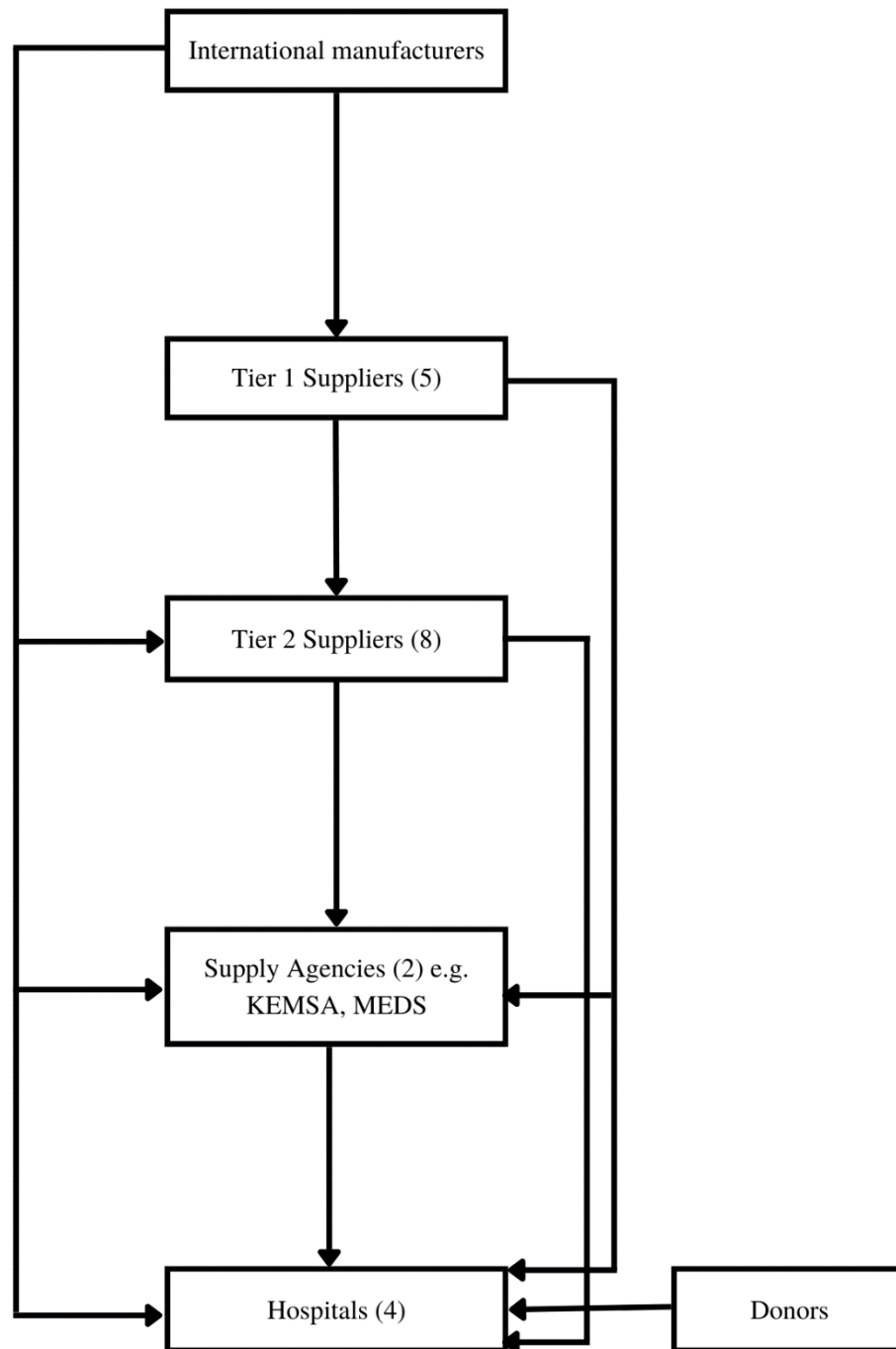
#### 4.1.1 *Types of Providers Surveyed*

Three main providers of anticancer drugs were surveyed during this study, namely, supply agencies, hospitals and suppliers. Supply agencies are large institutions with warehousing capability and countrywide supply chain whose main purpose is to supply drugs to hospitals at subsidized rates whereas suppliers are wholesalers and retailers who sell drugs commercially to diverse clientele. The three providers are related as illustrated in the flowchart (Figure 4.1). As illustrated, all providers, regardless of the tier, usually import some products from international manufacturers. Tier 1 suppliers, with high trade volumes import the bulk of the products and supply most of the other providers. Donors play a key role in the supply chain by providing drugs free of charge or subsidizing costs.

The largest supply agencies in Kenya, KEMSA and MEDS, were included in this survey. KEMSA mainly supplies public hospitals while MEDS mostly private hospitals and faith based organizations.

Four hospitals were purposively sampled because they are the key centers for cancer care in Nairobi. One of the hospitals was Kenyatta National Hospital (KNH), which is the largest hospital and mainly serves low and middle-income patients in the public sector. The other three facilities are the largest private hospitals in Kenya and mainly serve high-income patients.

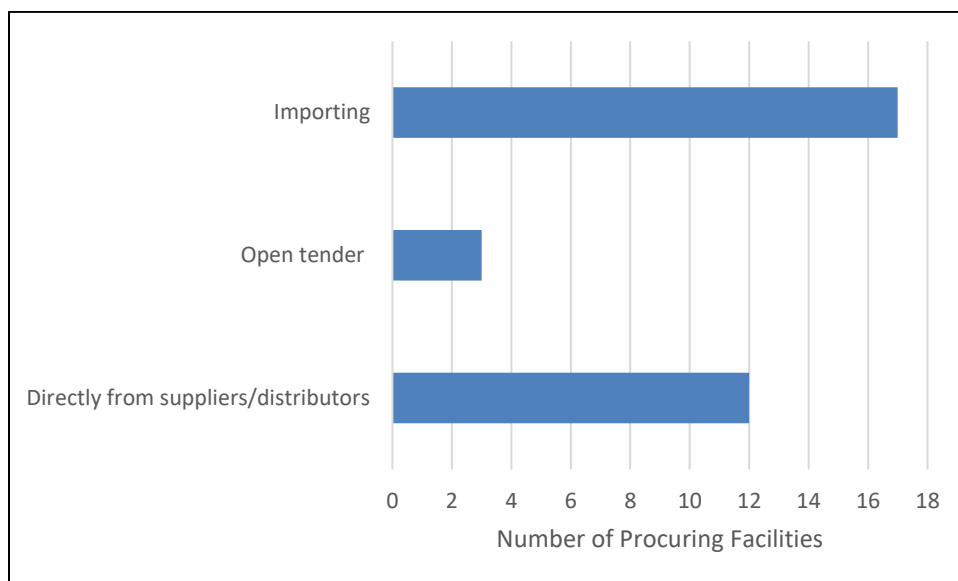
Suppliers included wholesalers and retailers. They were divided into two groups based on their trade volumes. The first tier of suppliers was made of 5, while the second tier was made up of 8.



*Figure 4.1 Relationship between the different providers of anticancer drugs in Nairobi, Kenya*

#### ***4.1.2 Procurement methods for anticancer drugs***

Three main methods for procurement of anticancer drugs were used, namely; direct purchase from suppliers, open tenders and direct importation. Most of the institutions used a mix of these methods to obtain their drugs while a few, especially government institutions used open tendering. As illustrated in Figure 4.2, most providers relied on importing their own drugs while the remaining products were obtained from local suppliers. Open tendering was used by KNH, KEMSA and MEDS.



*Figure 4.2 Procurement Methods Used to Acquire Anticancer Drugs*

## 4.2 Characteristics of The Surveyed Drugs

A total of 92 anticancer products were surveyed and the three main dosage forms encountered were injections, tablets and capsules. Injections represented the majority at (66, 71.7%) with tablets and capsules making up the rest (26, 28.3%).

Ten classes of drugs were surveyed. Molecular targeted agents also known as biologics were the majority (25, 27.2%) followed by alkylating agents (23, 25%) and plant derived agents (17, 18.5%).

The list of drugs to be surveyed had 50 anticancer drugs. However, these drugs exist in different strengths leading to a final list of 92 products. Paclitaxel and cyclophosphamide had four different formulations, while cisplatin, cytarabine, docetaxel, melphalan and trastuzumab had three each. Table 4.1 summarizes the dosages, classes and formulations of the surveyed drugs.

*Table 4.1 Dosage forms, Anticancer Classes and Formulations*

| <b>Dosage Form</b>                             | <b>Frequency</b> | <b>Percentage</b> |
|------------------------------------------------|------------------|-------------------|
| Injection                                      | 66               | 71.7              |
| Tablet/Capsule                                 | 26               | 28.3              |
| <b>Anticancer Class</b>                        | <b>Frequency</b> | <b>Percentage</b> |
| Enzymes                                        | 1                | 1.1               |
| Alkylating Agents                              | 23               | 25                |
| Antitumour Antibiotics                         | 8                | 8.7               |
| Detoxifying Agents                             | 2                | 2.2               |
| Antimetabolites                                | 14               | 15.2              |
| Plant Derived Agents                           | 17               | 18.5              |
| Retinoids                                      | 1                | 1.1               |
| Molecular Targeted Agents                      | 25               | 27.2              |
| Other Antineoplastic Agents (Arsenic Trioxide) | 1                | 1.1               |
| <b>Number of Formulations</b>                  | <b>Frequency</b> | <b>Percentage</b> |
| 1                                              | 17               | 34                |
| 2                                              | 26               | 52                |
| 3                                              | 5                | 10                |
| 4                                              | 2                | 4                 |

### **4.3 Availability of Anticancer Drugs At The Surveyed Providers**

For the supply agencies, KEMSA stocked 41 (44.6%) of the 92 drug products surveyed, while MEDS had 13 (14.1%). Two private hospitals had a higher availability of the surveyed anticancer drugs 59 (64.1%) and 54 (58.7%), while KNH 42 (45.7%), was third. The final private hospital had 32 (34.8%) of the anticancer drugs.

Five of the 13 suppliers surveyed had more than half of the target products surveyed (51.1-73.9%). One of the suppliers had as low as 3 (3.3%) products. Table 4.2 further illustrates the availability of the drugs in the different tiers of suppliers. It was observed that Tier 2 suppliers (40.2%) had a higher average availability of drugs than tier 1 suppliers (33.5%). This can be attributed to the fact that tier 2 suppliers are more open to stocking generic versions of all anticancer drugs while tier 1 suppliers mostly stock originator brands due to long standing relationships with multinational pharmaceutical companies.

Availability of anticancer drugs was highest at the hospitals (50.8%) followed by suppliers (37.6%) and then the supply agencies (29.4%). This can be explained by the fact that agencies do not place an emphasis on stocking of anticancer drugs because most of their main customers, public and mission hospitals, do not offer anticancer treatment.

Alkylating, antimetabolite, plant derived agents and antitumour antibiotics were available evenly across all the providers. However, molecular targeted agents were more available at suppliers and hospitals and least available at the agencies. This points to a slow adoption of stocking targeted therapies by agencies.

Appendix D shows availability of the different drugs by class across the surveyed providers. The most available drug was carboplatin 450mg and the least available drugs included thalidomide 50mg, cisplatin 100mg, melphalan 5mg, daunorubicin 10mg, vincristine 2mg and all trans retinoic acid 10mg. Arsenic trioxide and tioguanine were completely unavailable at all the surveyed providers.

Table 4.2 Availability at Agencies, Hospitals and Supply Agencies

| <b>Agency</b>  | <b>Frequency</b> | <b>Percentage</b> |
|----------------|------------------|-------------------|
| KEMSA          | 41               | 44.6              |
| MEDS           | 13               | 14.1              |
| <b>Average</b> |                  | <b>29.4</b>       |

| <b>Hospital</b>    | <b>Frequency</b> | <b>Percentage</b> |
|--------------------|------------------|-------------------|
| KNH                | 42               | 45.7              |
| Private Hospital 1 | 32               | 34.8              |
| Private Hospital 2 | 54               | 58.7              |
| Private Hospital 3 | 59               | 64.1              |
| <b>Average</b>     |                  | <b>50.8</b>       |

| <b>Tier 1 Suppliers</b> | <b>Frequency</b> | <b>Percentage</b> |
|-------------------------|------------------|-------------------|
| Supplier 1              | 47               | 51.1              |
| Supplier 2              | 12               | 13                |
| Supplier 3              | 52               | 56.5              |
| Supplier 4              | 29               | 31.5              |
| Supplier 5              | 14               | 15.2              |
| <b>Average</b>          |                  | <b>33.5</b>       |

| <b>Tier 2 Suppliers</b> | <b>Frequency</b> | <b>Percentage</b> |
|-------------------------|------------------|-------------------|
| Supplier 6              | 68               | 73.9              |
| Supplier 7              | 10               | 10.9              |
| Supplier 8              | 35               | 38                |
| Supplier 9              | 53               | 57.6              |
| Supplier 10             | 3                | 3.3               |
| Supplier 11             | 43               | 46.7              |
| Supplier 12             | 36               | 39.1              |
| Supplier 13             | 48               | 52.2              |
| <b>Average</b>          |                  | <b>40.2</b>       |

#### 4.4 Availability of Originator Brands and Lowest Priced Generics

From the list of anticancer drugs surveyed, there was a mix of originator brands and generics from the different service providers. Most suppliers stocked generics over originator brands while some stocked both types. This was the case for hospitals as well. Figure 4.3 illustrates the average availability of originator brands (OBs) and lowest price generics (LPGs) for the surveyed drugs. Appendix D shows availability of the complete list of originator brands and lowest priced generics for all the drugs.

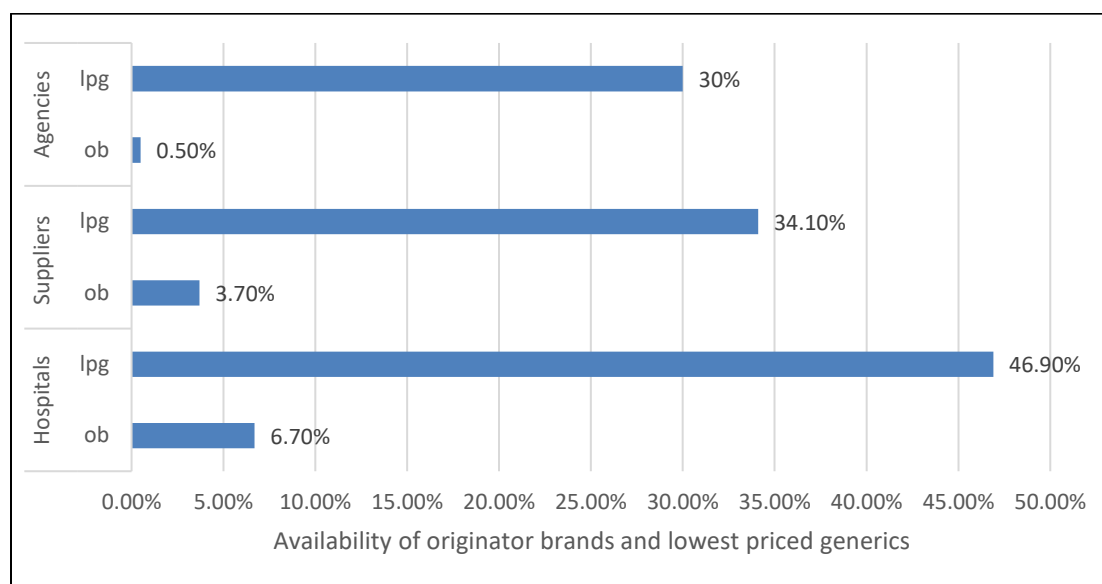


Figure 4.3 Bar graph showing Availability of Originator Brands and Lowest Priced Generics

#### 4.5 Prices of Anticancer Drugs At The Surveyed Providers

Prices of the different anticancer drugs were obtained during the survey and analysis conducted as follows. Maximum and minimum prices were obtained for all surveyed drugs, from all providers surveyed and price variation analysis was conducted.



#### 4.5.1 Price Variation Analysis of Molecular Targeted Agents, Enzymes and Other Antineoplastics

Imatinib 400mg had the highest price ratio of 15.5, which means that the highest market price was 15.5 times higher than its minimum price. Additionally, it had the highest price variation of 1454.3%. Thalidomide 50mg had the lowest price ratio of 1, which means that the price did not vary at all and was consistent across different suppliers.

Table 4.3 Price Variation Analysis for Molecular Targeted Agents, Enzymes and Other Antineoplastics

| Medicine Name    | Strength | Minimum Price (KES) | Maximum Price (KES) | Price Ratio | Price Variation % |
|------------------|----------|---------------------|---------------------|-------------|-------------------|
| Imatinib         | 400mg    | 105                 | 1632                | 15.5        | 1454.3            |
| Bortezomib       | 3.5mg    | 8000                | 91958               | 11.5        | 1049.5            |
| Gefitinib        | 250mg    | 113                 | 600                 | 5.3         | 431               |
| Nilotinib        | 200mg    | 260                 | 1188                | 4.6         | 356.9             |
| Bevacizumab      | 100mg    | 15000               | 67000               | 4.5         | 346.7             |
| Trastuzumab      | 440mg    | 45000               | 188232              | 4.2         | 318.3             |
| Lenalidomide     | 10mg     | 150                 | 600                 | 4           | 300               |
| Rituximab        | 500mg    | 42000               | 164487              | 3.9         | 291.6             |
| Bevacizumab      | 400mg    | 50000               | 180000              | 3.6         | 260               |
| Bortezomib       | 2mg      | 5740                | 20000               | 3.5         | 248.4             |
| Lenalidomide     | 25mg     | 367                 | 1250                | 3.4         | 240.6             |
| Thalidomide      | 100mg    | 100                 | 330                 | 3.3         | 230               |
| Rituximab        | 100mg    | 10450               | 34377               | 3.3         | 229               |
| Trastuzumab      | 150mg    | 20900               | 67226               | 3.2         | 221.7             |
| Pertuzumab       | 420mg    | 201431              | 610000              | 3           | 202.8             |
| Pembrolizumab    | 100mg    | 176000              | 520000              | 3           | 195.5             |
| Cetuximab        | 100mg    | 20645               | 51635               | 2.5         | 150.1             |
| Cetuximab        | 500mg    | 91350               | 228390              | 2.5         | 150               |
| Trastuzumab      | 600mg    | 76667               | 178966              | 2.3         | 133.4             |
| Erlotinib        | 150mg    | 400                 | 930                 | 2.3         | 132.5             |
| Imatinib         | 100mg    | 66                  | 140                 | 2.1         | 112.1             |
| Asparaginase     | 10.000IU | 3000                | 5520                | 1.8         | 84                |
| Filgastrim       | 300mcg   | 1200                | 1961                | 1.6         | 63.4              |
| Nilotinib        | 150mg    | 795                 | 1108                | 1.4         | 39.4              |
| Sorafenib        | 200mg    | 150                 | 208                 | 1.4         | 38.7              |
| Thalidomide      | 50mg     | 60                  | 60                  | 1           | 0                 |
| Arsenic Trioxide | 1mg      | NA                  | NA                  |             |                   |

From Table 4.3, we can observe that apart from imatinib 400mg, bortezomib 3.5mg and gefitinib 250mg, the price ratios were less than 5. This can be attributed to few biosimilars for most of the biological products in the market.

#### **4.5.2 Price Variation Analysis for Alkylating Agents, Detoxifying Agents and Antimetabolites**

Mercaptopurine had the highest price ratio of 15.7 and the highest price variation of 1471.4%. Cisplatin 100mg had the lowest price ratio.

*Table 4.4 Price Variation Analysis for Alkylating Agents, Detoxifying Agents and Antimetabolites*

| <b>Medicine Name</b> | <b>Strength</b> | <b>Minimum Price (KES)</b> | <b>Maximum Price (KES)</b> | <b>Price Ratio</b> | <b>Price variation %</b> |
|----------------------|-----------------|----------------------------|----------------------------|--------------------|--------------------------|
| Tioguanine           | 40mg            | NA                         | NA                         |                    | NA                       |
| Mercaptopurine       | 50mg            | 14                         | 220                        | 15.7               | 1471.4                   |
| Cyclophosphamide     | 500mg           | 150                        | 2310                       | 15.4               | 1440.0                   |
| Cytarabine           | 1000mg          | 300                        | 4500                       | 15.0               | 1400.0                   |
| Hydroxycarbamide     | 500mg           | 19                         | 236                        | 12.4               | 1142.1                   |
| Fluorouracil         | 500mg           | 75                         | 787                        | 10.5               | 949.3                    |
| Capecitabine         | 500mg           | 92                         | 900                        | 9.8                | 878.3                    |
| Melphalan            | 2mg             | 102                        | 980                        | 9.6                | 860.8                    |
| Fluorouracil         | 250mg           | 45                         | 400                        | 8.9                | 788.9                    |
| Calcium folinate     | 15mg            | 17.5                       | 150                        | 8.6                | 757.1                    |
| Chlorambucil         | 2mg             | 68                         | 500                        | 7.4                | 635.3                    |
| Cyclophosphamide     | 1000mg          | 230                        | 1600                       | 7.0                | 595.7                    |
| Gemcitabine          | 200mg           | 538                        | 3500                       | 6.5                | 550.6                    |
| Cyclophosphamide     | 200mg           | 150                        | 972.8                      | 6.5                | 548.5                    |
| Methotrexate         | 50mg            | 140                        | 800                        | 5.7                | 471.4                    |
| Cisplatin            | 10mg            | 190                        | 960                        | 5.1                | 405.3                    |
| Bendamustine         | 100mg           | 6895                       | 29500                      | 4.3                | 327.8                    |
| Oxaliplatin          | 100mg           | 1800                       | 7680                       | 4.3                | 326.7                    |
| Gemcitabine          | 1000mg          | 1656                       | 7000                       | 4.2                | 322.7                    |
| Ifosfamide + mesna   | 1000mg + 600mg  | 685                        | 2650                       | 3.9                | 286.9                    |
| Cisplatin            | 50mg            | 490                        | 1868                       | 3.8                | 281.2                    |
| Oxaliplatin          | 50mg            | 1235                       | 4352                       | 3.5                | 252.4                    |

Table 4.4 Continued

| Medicine Name      | Strength        | Minimum Price (KES) | Maximum Price (KES) | Price Ratio | Price variation % |
|--------------------|-----------------|---------------------|---------------------|-------------|-------------------|
| Dacarbazine        | 200mg           | 600                 | 2000                | 3.3         | 233.3             |
| Capecitabine       | 150mg           | 38                  | 120                 | 3.2         | 215.8             |
| Cytarabine         | 100mg           | 265                 | 835.58              | 3.2         | 215.3             |
| Cytarabine         | 500mg           | 1000                | 3100                | 3.1         | 210.0             |
| Methotrexate       | 2.5mg           | 6                   | 17                  | 2.8         | 183.3             |
| Bendamustine       | 25mg            | 2750                | 7700                | 2.8         | 180.0             |
| Carboplatin        | 150mg           | 1350                | 2560                | 1.9         | 89.6              |
| Procarbazine       | 50mg            | 80                  | 150                 | 1.9         | 87.5              |
| Carboplatin        | 450mg           | 3250                | 6000                | 1.8         | 84.6              |
| Ifosfamide + mesna | 2000mg + 1200mg | 1370                | 2460                | 1.8         | 79.6              |
| Calcium folinate   | 50mg            | 380                 | 600                 | 1.6         | 57.9              |
| Dacarbazine        | 500mg           | 1425                | 1900                | 1.3         | 33.3              |
| Chlorambucil       | 5mg             | 201                 | 250                 | 1.2         | 24.4              |
| Cyclophosphamide   | 50mg            | 60                  | 64.77               | 1.1         | 7.9               |
| Cisplatin          | 100mg           | 2000                | 2000                | 1.0         | 0.0               |
| Melphalan          | 5mg             | 233                 | 233                 | 1.0         | 0.0               |
| Melphalan          | 50mg            | 7474                | 7474                | 1.0         | 0.0               |

From Table 4.4, it can be observed that unlike in the case of biologics, there were 15 products, which had a price ratios above 5. This can be attributed to a higher availability of generic products for alkylating and antimetabolite agents.

### 4.5.3 Price Variation Analysis for Plant Derived agents, Antitumor antibiotics and Retinoids

Table 4.5 Price Analysis for Plant Derived agents, Antitumor antibiotics and Retinoids

| Medicine Name           | Strength | Minimum Price (KES) | Maximum Price (KES) | Price Ratio | Price variation % |
|-------------------------|----------|---------------------|---------------------|-------------|-------------------|
| Docetaxel               | 20mg     | 950                 | 14300               | 15.1        | 1405.3            |
| Doxorubicin             | 50mg     | 450                 | 6600                | 14.7        | 1366.7            |
| Doxorubicin             | 10mg     | 250                 | 1800                | 7.2         | 620.0             |
| Vinorelbine             | 50mg     | 6072                | 42000               | 6.9         | 591.7             |
| Vinblastine             | 10mg     | 600                 | 4000                | 6.7         | 566.7             |
| Docetaxel               | 80mg     | 2400                | 12082.33            | 5.0         | 403.4             |
| Paclitaxel              | 300mg    | 2900                | 14528               | 5.0         | 401.0             |
| Etoposide               | 50mg     | 80                  | 400                 | 5.0         | 400.0             |
| Liposomal Doxorubicin   | 20mg     | 7500                | 33439               | 4.5         | 345.9             |
| Irinotecan              | 40mg     | 1116                | 4750                | 4.3         | 325.6             |
| Daunorubicin            | 20mg     | 900                 | 3800                | 4.2         | 322.2             |
| Vincristine             | 1mg      | 140                 | 550                 | 3.9         | 292.9             |
| Paclitaxel              | 100mg    | 1200                | 4600                | 3.8         | 283.3             |
| Docetaxel               | 120mg    | 2690                | 9600                | 3.6         | 256.9             |
| Irinotecan              | 100mg    | 1800                | 4875                | 2.7         | 170.8             |
| Etoposide               | 100mg    | 320                 | 774                 | 2.4         | 141.9             |
| Vinorelbine             | 10mg     | 3200                | 7590                | 2.4         | 137.2             |
| Liposomal Doxorubicin   | 50mg     | 16000               | 35200               | 2.2         | 120.0             |
| Paclitaxel              | 30mg     | 785                 | 1650                | 2.1         | 110.2             |
| Bleomycin               | 15iu     | 1000                | 2100                | 2.1         | 110.0             |
| Dactinomycin            | 0.5mg    | 850                 | 1715.2              | 2.0         | 101.8             |
| Topotecan               | 2.5mg    | 10450               | 18600               | 1.8         | 78.0              |
| Paclitaxel              | 260mg    | 3800                | 5500                | 1.4         | 44.7              |
| Daunorubicin            | 10mg     | 1088                | 1088                | 1.0         | 0.0               |
| Vincristine             | 2mg      | 950                 | 950                 | 1.0         | 0.0               |
| All trans retinoic acid | 10mg     | 1494                | 1494                | 1.0         | 0.0               |

Docetaxel 20mg had the highest price ratio of 15.1 whereas Daunorubicin 10mg had the lowest price ratio. From Table 4.5, we can also observe that five drugs had price ratios above 5. This points to a lower availability of generics of anticancer drugs derived from natural sources, as compared to alkylating and antimetabolite agents.

#### ***4.5.4 Comparison of Prices of Selected Anticancer Agents at the Supply Agencies***

Out of all the drugs, only 8 drugs were available in both KEMSA and MEDS. The prices of these 8 products were compared and the findings are presented in Table 4.6. Of the 8 products, KEMSA had 5 drugs selling at higher prices compared to MEDS. The expectations is that KEMSA should have the lower prices for the drugs but this was not the case. The higher prices can be attributed to inflated prices during the tendering process. Price ratios were calculated by dividing the KEMSA prices by MEDS prices.

*Table 4.6 Price Analysis at the Supply Agencies*

| <b>Medicine</b>  | <b>Strength</b> | <b>KEMSA<br/>(KES)</b> | <b>MEDS (KES)</b> | <b>Price Ratio</b> |
|------------------|-----------------|------------------------|-------------------|--------------------|
| Cytarabine       | 100mg           | 265                    | 466               | 0.6                |
| Doxorubicin      | 50mg            | 450                    | 552               | 0.8                |
| Cyclophosphamide | 200mg           | 220                    | 224               | 1.0                |
| Gemcitabine      | 1000mg          | 1743                   | 1656              | 1.1                |
| Gemcitabine      | 200mg           | 592                    | 538               | 1.1                |
| Vinblastine      | 10mg            | 1140                   | 748               | 1.5                |
| Daunorubicin     | 20mg            | 1575                   | 978               | 1.6                |
| Mercaptopurine   | 50mg            | 220                    | 27                | 8.1                |

#### ***4.5.5 Comparison of Prices of Selected Anticancer Agents at the Hospitals***

Price ratios were generated by establishing the maximum and minimum prices for the available drugs from all the hospitals. Using drugs available in KNH as the baseline, comparisons of the prices were made against the lowest priced drug in the three private hospitals. KNH prices were used as the denominators in calculation of the price ratios; the assumption being that it should provide the cheapest alternative being a public institution. Of the 40 drugs found in KNH and the 3 private hospitals, 19 of them were available in the private sector at cheaper prices. Highlighted in Table 4.7 are 10 products with their price ratios. KNH uses open tendering process for procurement of drug supplies and inflated prices during this process might lead to these differences. In

addition, some private hospitals import drugs for themselves therefore, their prices might be lower as a result of this.

*Table 4.7 List of drugs that are more expensive in KNH*

| <b>Medicine Name</b>  | <b>Strength</b> | <b>KNH (KES)</b> | <b>Lowest Priced Private Hospital (KES)</b> | <b>Price Ratio</b> |
|-----------------------|-----------------|------------------|---------------------------------------------|--------------------|
| Melphalan             | 2mg             | 980              | 200                                         | 0.2                |
| Bevacizumab           | 100mg           | 50000            | 18216                                       | 0.4                |
| Gefitinib             | 250mg           | 260              | 113                                         | 0.4                |
| Trastuzumab           | 440mg           | 98000            | 56400                                       | 0.6                |
| Liposomal Doxorubicin | 20mg            | 24000            | 14050                                       | 0.6                |
| Irinotecan            | 100mg           | 4875             | 2900                                        | 0.6                |
| Dactinomycin          | 0.5mg           | 1430             | 900                                         | 0.6                |
| Capecitabine          | 500mg           | 190              | 120                                         | 0.6                |
| Rituximab             | 500mg           | 111000           | 73600                                       | 0.7                |
| Calcium folinate      | 15mg            | 140              | 100                                         | 0.7                |

That notwithstanding, KNH had the lower prices for 20 of the 40 drugs listed in Table 4.8.

*Table 4.8 List of drugs that are cheaper at KNH*

| <b>Medicine Name</b> | <b>Strength</b> | <b>KNH (KES)</b> | <b>Lowest Priced Private Hospital (KES)</b> | <b>Price Ratio</b> |
|----------------------|-----------------|------------------|---------------------------------------------|--------------------|
| Bendamustine         | 100mg           | 9620             | 10200                                       | 1.1                |
| Dacarbazine          | 200mg           | 960              | 1030                                        | 1.1                |
| Ifosfamide + mesna   | 1000mg + 600mg  | 1700             | 1920                                        | 1.1                |
| Filgastrim           | 300mcg          | 1235             | 1400                                        | 1.1                |
| Thalidomide          | 100mg           | 110              | 128                                         | 1.2                |
| Cisplatin            | 50mg            | 640              | 793.6                                       | 1.2                |
| Daunorubicin         | 20mg            | 1100             | 1400                                        | 1.3                |
| Chlorambucil         | 2mg             | 200              | 278.27                                      | 1.4                |
| Carboplatin          | 450mg           | 3650             | 5376                                        | 1.5                |
| Etoposide            | 100mg           | 350              | 550                                         | 1.6                |
| Oxaliplatin          | 100mg           | 2400             | 3900                                        | 1.6                |
| Procarbazine         | 50mg            | 80               | 133                                         | 1.7                |
| Vinblastine          | 10mg            | 600              | 1050                                        | 1.8                |
| Hydroxycarbamide     | 500mg           | 25               | 47                                          | 1.9                |
| Gemcitabine          | 200mg           | 980              | 1920                                        | 2.0                |
| Mercaptopurine       | 50mg            | 25               | 50                                          | 2.0                |
| Paclitaxel           | 100mg           | 1550             | 4480                                        | 2.9                |

|                  |       |     |      |     |
|------------------|-------|-----|------|-----|
| Lenalidomide     | 10mg  | 150 | 448  | 3.0 |
| Nilotinib        | 200mg | 260 | 1186 | 4.6 |
| Cyclophosphamide | 500mg | 250 | 1152 | 4.6 |

#### ***4.5.5.1 Variability in Prices of Selected Anticancer Agents across the Hospitals***

Listed in Table 4.9 are 5 of the lowest and highest price ratios out of the products available in hospitals. Fluorouracil 500mg had the highest price ratio at 8.2, followed by mercaptopurine and plant derived agents such as docetaxel and etoposide. This can be attributed to availability of differently priced brands that the different hospitals choose to stock. In contrast, molecular agents such as pembrolizumab and sorafenib had low price ratios because of unavailability of low priced biosimilars.

*Table 4.9 Variability of prices across hospitals*

| <b>Medicine Name</b> | <b>Strength</b> | <b>Minimum Price (KES)</b> | <b>Maximum Price (KES)</b> | <b>Price Ratio</b> |
|----------------------|-----------------|----------------------------|----------------------------|--------------------|
| Trastuzumab          | 600mg           | 86432                      | 87630                      | 1.0                |
| Bleomycin            | 15iu            | 1500                       | 1529.6                     | 1.0                |
| Cytarabine           | 100mg           | 800                        | 835.58                     | 1.0                |
| Sorafenib            | 200mg           | 150                        | 158                        | 1.1                |
| Pembrolizumab        | 100mg           | 211200                     | 225280                     | 1.1                |
| Melphalan            | 2mg             | 200                        | 980                        | 4.9                |
| Etoposide            | 50mg            | 80                         | 400                        | 5.0                |
| Docetaxel            | 80mg            | 2400                       | 12082.33                   | 5.0                |
| Mercaptopurine       | 50mg            | 25                         | 155                        | 6.2                |
| Fluorouracil         | 500mg           | 96                         | 787                        | 8.2                |

#### ***4.5.6 Comparison of Prices of Selected Anticancer Agents Across the Suppliers***

Price analysis was conducted by obtaining the minimum and maximum prices for available anticancer drugs from the suppliers, and calculating the resultant price ratios. Listed in Table 4.10, are the five lowest and highest price ratios from the suppliers. It was observed that drugs which are rarely used had low price ratios while the widely used ones had higher price ratios. The large price ratios (11.5 – 15.4) for bortezomib to

cyclophosphamide, can be attributed to the different suppliers stocking originator brands vis a vis lower priced generics.

*Table 4.10 Price Ratios from Suppliers*

| <b>Medicine Name</b> | <b>Strength</b> | <b>Minimum Price (KES)</b> | <b>Maximum Price (KES)</b> | <b>Price Ratio</b> |
|----------------------|-----------------|----------------------------|----------------------------|--------------------|
| Calcium folinate     | 50mg            | 380                        | 400                        | 1.1                |
| Thalidomide          | 100mg           | 100                        | 110                        | 1.1                |
| Nilotinib            | 200mg           | 1033                       | 1188                       | 1.2                |
| Capecitabine         | 150mg           | 75                         | 89                         | 1.2                |
| Procarbazine         | 50mg            | 85                         | 104                        | 1.2                |
| Bortezomib           | 3.5mg           | 8000                       | 91958                      | 11.5               |
| Hydroxycarbamide     | 500mg           | 19                         | 236                        | 12.4               |
| Cytarabine           | 1000mg          | 300                        | 4500                       | 15.0               |
| Docetaxel            | 20mg            | 950                        | 14300                      | 15.1               |
| Cyclophosphamide     | 500mg           | 150                        | 2310                       | 15.4               |

#### **4.5.7 Price Margin Analysis**

Since suppliers and agencies are the largest source of anticancer drugs, analysis was done to establish the margins that are added to determine the final price for the consumers.

Firstly, KEMSA was assumed to be the primary source of drugs and percentage margins calculated as shown in Table 4.11. It was interesting to note that KEMSA being a supply agency, had 4 products at a higher price than KNH which sells directly to the consumer. This can be attributed to the procurement practices at KEMSA that lead to inflated prices. Drugs with the highest margins include calcium folinate, methotrexate and chlorambucil whereas those with the lowest margins include carboplatin, etoposide and trastuzumab.



Table 4.11 Price Margin analysis of KEMSA and KNH

| Medicine Name      | Strength          | KNH Price<br>(KES) | KEMSA Price<br>(KES) | Margin | Percent<br>Margin |
|--------------------|-------------------|--------------------|----------------------|--------|-------------------|
| Mercaptopurine     | 50mg              | 25                 | 220                  | -195   | -88.6             |
| Vinblastine        | 10mg              | 600                | 1140                 | -540   | -47.4             |
| Daunorubicin       | 20mg              | 1100               | 1575                 | -475   | -30.2             |
| Filgastrim         | 300mcg            | 1235               | 1400                 | -165   | -11.8             |
| Bleomycin          | 15iu              | 1500               | 1500                 | 0      | 0.0               |
| Carboplatin        | 450mg             | 3650               | 3500                 | 150    | 4.3               |
| Etoposide          | 100mg             | 350                | 320                  | 30     | 9.4               |
| Trastuzumab        | 440mg             | 98000              | 88995                | 9005   | 10.1              |
| Rituximab          | 500mg             | 111000             | 98550                | 12450  | 12.6              |
| Hydroxycarbamide   | 500mg             | 25                 | 22                   | 3      | 13.6              |
| Vincristine        | 1mg               | 360                | 305                  | 55     | 18.0              |
| Capecitabine       | 500mg             | 190                | 154                  | 36     | 23.4              |
| Paclitaxel         | 100mg             | 1550               | 1200                 | 350    | 29.2              |
| Dacarbazine        | 200mg             | 960                | 740                  | 220    | 29.7              |
| Dactinomycin       | 0.5mg             | 1430               | 1098                 | 332    | 30.2              |
| Cisplatin          | 50mg              | 640                | 490                  | 150    | 30.6              |
| Oxaliplatin        | 100mg             | 2400               | 1800                 | 600    | 33.3              |
| Cyclophosphamide   | 500mg             | 250                | 185                  | 65     | 35.1              |
| Fluorouracil       | 250mg             | 90                 | 63                   | 27     | 42.9              |
| Bortezomib         | 3.5mg             | 12350              | 8630                 | 3720   | 43.1              |
| Asparaginase       | 10,000<br>IU      | 5135               | 3300                 | 1835   | 55.6              |
| Gemcitabine        | 200mg             | 980                | 592                  | 388    | 65.5              |
| Doxorubicin        | 50mg              | 860                | 450                  | 410    | 91.1              |
| Chlorambucil       | 2mg               | 200                | 89                   | 111    | 124.7             |
| Methotrexate       | 2.5mg             | 17                 | 7                    | 10     | 142.9             |
| Ifosfamide + mesna | 1000mg<br>+ 600mg | 1700               | 685                  | 1015   | 148.2             |
| Calcium folinate   | 15mg              | 140                | 17.5                 | 122.5  | 700.0             |

Secondly, the lowest prices from the suppliers were used to calculate the margins if KNH was to obtain these drugs from suppliers. As was observed with KEMSA, the first 8 drugs listed in Table 4.12 were cheaper in KNH, than the suppliers. This can be attributed to the open tendering process.

Table 4.12 Price Margin analysis of KNH and Lowest Prices from Suppliers

| Medicine Name      | Strength       | KNH Price (KES) | Lowest<br>Priced<br>Supplier<br>(KES) | Margin | Percent<br>Margin |
|--------------------|----------------|-----------------|---------------------------------------|--------|-------------------|
| Nilotinib          | 200mg          | 260             | 1033                                  | -773   | -74.8             |
| Lenalidomide       | 10mg           | 150             | 288                                   | -138   | -47.9             |
| Vinblastine        | 10mg           | 600             | 850                                   | -250   | -29.4             |
| Etoposide          | 100mg          | 350             | 450                                   | -100   | -22.2             |
| Gemcitabine        | 200mg          | 980             | 1100                                  | -120   | -10.9             |
| Calcium folinate   | 15mg           | 140             | 150                                   | -10    | -6.7              |
| Procarbazine       | 50mg           | 80              | 85                                    | -5     | -5.9              |
| Topotecan          | 2.5mg          | 10450           | 11000                                 | -550   | -5.0              |
| Filgastrim         | 300mcg         | 1235            | 1200                                  | 35     | 2.9               |
| Paclitaxel         | 100mg          | 1550            | 1500                                  | 50     | 3.3               |
| Docetaxel          | 80mg           | 2640            | 2500                                  | 140    | 5.6               |
| Thalidomide        | 100mg          | 110             | 100                                   | 10     | 10.0              |
| Cisplatin          | 50mg           | 640             | 580                                   | 60     | 10.3              |
| Carboplatin        | 450mg          | 3650            | 3250                                  | 400    | 12.3              |
| Oxaliplatin        | 100mg          | 2400            | 2090                                  | 310    | 14.8              |
| Gefitinib          | 250mg          | 260             | 218                                   | 42     | 19.3              |
| Daunorubicin       | 20mg           | 1100            | 900                                   | 200    | 22.2              |
| Cetuximab          | 100mg          | 25350           | 20645                                 | 4705   | 22.8              |
| Hydroxycarbamide   | 500mg          | 25              | 19                                    | 6      | 31.6              |
| Bendamustine       | 100mg          | 9620            | 6895                                  | 2725   | 39.5              |
| Vinorelbine        | 50mg           | 15400           | 11000                                 | 4400   | 40.0              |
| Doxorubicin        | 50mg           | 860             | 600                                   | 260    | 43.3              |
| Ifosfamide + mesna | 1000mg + 600mg | 1700            | 1140                                  | 560    | 49.1              |
| Bleomycin          | 15iu           | 1500            | 1000                                  | 500    | 50.0              |
| Bortezomib         | 3.5mg          | 12350           | 8000                                  | 4350   | 54.4              |
| Dacarbazine        | 200mg          | 960             | 600                                   | 360    | 60.0              |
| Cyclophosphamide   | 500mg          | 250             | 150                                   | 100    | 66.7              |
| Dactinomycin       | 0.5mg          | 1430            | 850                                   | 580    | 68.2              |
| Asparaginase       | 10,000 IU      | 5135            | 3000                                  | 2135   | 71.2              |
| Mercaptopurine     | 50mg           | 25              | 14                                    | 11     | 78.6              |
| Fluorouracil       | 250mg          | 90              | 45                                    | 45     | 100.0             |
| Capecitabine       | 500mg          | 190             | 92                                    | 98     | 106.5             |
| Trastuzumab        | 440mg          | 98000           | 45000                                 | 53000  | 117.8             |
| Vincristine        | 1mg            | 360             | 140                                   | 220    | 157.1             |
| Rituximab          | 500mg          | 111000          | 42000                                 | 69000  | 164.3             |
| Irinotecan         | 100mg          | 4875            | 1800                                  | 3075   | 170.8             |
| Methotrexate       | 2.5mg          | 17              | 6                                     | 11     | 183.3             |
| Chlorambucil       | 2mg            | 200             | 68                                    | 132    | 194.1             |

Table 4.12 Continued

| Medicine Name            | Strength | KNH Price (KES) | Lowest<br>Priced<br>Supplier<br>(KES) | Margin | Percent<br>Margin |
|--------------------------|----------|-----------------|---------------------------------------|--------|-------------------|
| Liposomal<br>Doxorubicin | 20mg     | 24000           | 7500                                  | 16500  | 220.0             |
| Bevacizumab              | 100mg    | 50000           | 15000                                 | 35000  | 233.3             |
| Cytarabine               | 1000mg   | 2840            | 300                                   | 2540   | 846.7             |
| Melphalan                | 2mg      | 980             | 102                                   | 878    | 860.8             |

## Price Margin Analysis of Anticancer drugs in Private Hospitals

Private Hospital 3, which had the highest availability, was also used to conduct margin analysis against KEMSA and the cheapest supplier prices. It was interesting to observe that 14 products had margins above 100%. However, the private hospital had 5 products selling at a cheaper price than KEMSA. This discrepancy can be attributed to differences in procurement practices and stocking of originator and generic products. Table 4.13 shows ten of the highest and lowest percent margins.

Table 4.13 Price Margin analysis of Private hospital 3 and KEMSA prices

| Medicine Name    | Strength  | Private Hospital 3<br>(KES) | KEMSA<br>Price (KES) | Margin   | Percent<br>Margin |
|------------------|-----------|-----------------------------|----------------------|----------|-------------------|
| Cyclophosphamide | 500mg     | 1152                        | 185                  | 967.0    | 522.7             |
| Cyclophosphamide | 1000mg    | 1382.4                      | 230                  | 1152.4   | 501.0             |
| Cyclophosphamide | 200mg     | 972.8                       | 220                  | 752.8    | 342.2             |
| Oxaliplatin      | 100mg     | 7680                        | 1800                 | 5880.0   | 326.7             |
| Paclitaxel       | 100mg     | 4480                        | 1200                 | 3280.0   | 273.3             |
| Gemcitabine      | 1000mg    | 5760                        | 1743                 | 4017.0   | 230.5             |
| Cytarabine       | 100mg     | 835.58                      | 265                  | 570.6    | 215.3             |
| Chlorambucil     | 2mg       | 278.27                      | 89                   | 189.3    | 212.7             |
| Oxaliplatin      | 50mg      | 4352                        | 1500                 | 2852.0   | 190.1             |
| Methotrexate     | 50mg      | 512                         | 180                  | 332.0    | 184.4             |
| Calcium folinate | 50mg      | 576                         | 440                  | 136.0    | 30.9              |
| Asparaginase     | 10,000 IU | 4160                        | 3300                 | 860.0    | 26.1              |
| Vincristine      | 1mg       | 384                         | 305                  | 79.0     | 25.9              |
| Rituximab        | 100mg     | 17664                       | 15000                | 2664.0   | 17.8              |
| Bleomycin        | 15iu      | 1529.6                      | 1500                 | 29.6     | 2.0               |
| Vinblastine      | 10mg      | 1088                        | 1140                 | -52.0    | -4.6              |
| Cytarabine       | 500mg     | 2643.2                      | 3100                 | -456.8   | -14.7             |
| Rituximab        | 500mg     | 73600                       | 98550                | -24950.0 | -25.3             |
| Trastuzumab      | 440mg     | 62720                       | 88995                | -26275.0 | -29.5             |
| Mercaptopurine   | 50mg      | 155                         | 220                  | -65.0    | -29.5             |

Price margins for a number of the drugs were very large, ranging from 184.4% - 522.7%. In Kenya, the agreed upon drug margins are 30%. This, therefore, shows cases of excess profits.

Price margin analysis was also conducted for private hospital 3 against the cheapest supplier price. The private hospital had 4 products cheaper than the all the suppliers. 22 of the products had margins above 100%. Table 4.14 shows the ten highest and lowest percentage margins.

*Table 4.14 Price Margin analysis for Private Hospital 3 against lowest prices from the suppliers*

| <b>Medicine Name</b> | <b>Strength</b> | <b>Private Hospital 3<br/>Price (KES)</b> | <b>Minimum<br/>Supplier price<br/>(KES)</b> | <b>Margin</b> | <b>Percent Margin</b> |
|----------------------|-----------------|-------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Mercaptopurine       | 50mg            | 155                                       | 14                                          | 141           | 1007.1                |
| Fluorouracil         | 500mg           | 787                                       | 75                                          | 712           | 949.3                 |
| Cyclophosphamide     | 500mg           | 1152                                      | 150                                         | 1002          | 668.0                 |
| Cyclophosphamide     | 200mg           | 972.8                                     | 150                                         | 822.8         | 548.5                 |
| Cyclophosphamide     | 1000mg          | 1382.4                                    | 250                                         | 1132.4        | 453.0                 |
| Paclitaxel           | 300mg           | 14528                                     | 2900                                        | 11628         | 401.0                 |
| Docetaxel            | 80mg            | 12082.33                                  | 2500                                        | 9582.33       | 383.3                 |
| Chlorambucil         | 2mg             | 278.27                                    | 68                                          | 210.27        | 309.2                 |
| Oxaliplatin          | 100mg           | 7680                                      | 2090                                        | 5590          | 267.5                 |
| Methotrexate         | 50mg            | 512                                       | 140                                         | 372           | 265.7                 |
| Thalidomide          | 100mg           | 128                                       | 100                                         | 28            | 28.0                  |
| Capecitabine         | 150mg           | 90.39                                     | 75                                          | 15.39         | 20.5                  |
| Dacarbazine          | 500mg           | 1715.2                                    | 1425                                        | 290.2         | 20.4                  |
| Cetuximab            | 100mg           | 24320                                     | 20645                                       | 3675          | 17.8                  |
| Nilotinib            | 200mg           | 1186                                      | 1033                                        | 153           | 14.8                  |
| Trastuzumab          | 150mg           | 23040                                     | 20900                                       | 2140          | 10.2                  |
| Bevacizumab          | 400mg           | 51200                                     | 52000                                       | -800          | -1.5                  |
| Pertuzumab           | 420mg           | 201431                                    | 206828                                      | -5396.96      | -2.6                  |
| Etoposide            | 50mg            | 80                                        | 130                                         | -50           | -38.5                 |
| Gefitinib            | 250mg           | 113                                       | 218                                         | -105          | -48.2                 |

#### **4.6 Comparison of Median Unit Prices Obtained From The Three Main Sources With International Reference Prices**

International Median Unit Prices were obtained from MSH International Price Guide 2015. The median unit prices from the three main sources were converted to US Dollars by using an exchange rate of 1 USD = 116.81 KES. This rate was obtained from the Central bank of Kenya website as the exchange rate for the month of June 2022, when data collection ended. The MSH guide did not include 15 of anticancer drugs obtained from the KEML. This disparity is attributed to the difference in years of publishing of the MSH Guide (2015), and the KEML (2019).

As can be observed in Table 4.15, median unit prices from the three providers were mostly comparable with the MSH prices and some were slightly cheaper. However, a few drugs had prices higher than MSH reference prices. The drugs included; cytarabine 500mg, docetaxel 20mg, docetaxel 80mg, doxorubicin 10mg, melphalan 50mg, vinblastine 10mg, vincristine 2mg, vinorelbine 50mg, vinorelbine 10mg, imatinib 100mg and nilotinib 200mg.

Table 4.15 Comparison of Median Unit Prices of Selected Anticancer Agents

| Medicine Name    | Strength | Dosage Form | MSH (USD) | Agencies (USD) | Hospitals (USD) | Suppliers (USD) |
|------------------|----------|-------------|-----------|----------------|-----------------|-----------------|
| Bleomycin        | 15iu     | injection   | 12.2      | 12.8           | 12.8            | 15.4            |
| Calcium folinate | 50mg     | injection   | 2.3       | 3.8            | 4.9             | 3.4             |
| Calcium folinate | 15mg     | tablet      | 1.3       | 0.1            | 1.0             | 1.3             |
| Capecitabine     | 150mg    | tablet      | 0.4       | 0.3            | 0.8             | 0.7             |
| Capecitabine     | 500mg    | tablet      | 1.7       | 1.3            | 1.8             | 1.5             |
| Carboplatin      | 150mg    | injection   | 14.9      | 11.6           | 21.9            | 15.4            |
| Carboplatin      | 450mg    | injection   | 34.3      | 30.0           | 48.7            | 33.4            |
| Chlorambucil     | 2mg      | tablet      | 0.7       | 0.8            | 2.4             | 1.0             |
| Cisplatin        | 50mg     | injection   | 6.0       | 4.2            | 7.2             | 6.4             |
| Cyclophosphamide | 50mg     | tablet      | 0.03      | -              | 0.5             | -               |
| Cyclophosphamide | 500mg    | injection   | 5.2       | 1.6            | 6.0             | 3.0             |
| Cyclophosphamide | 200mg    | injection   | 4.2       | 1.9            | 8.3             | 1.7             |
| Cytarabine       | 100mg    | injection   | 3.1       | 3.1            | 7.2             | 4.1             |
| Cytarabine       | 500mg    | injection   | 4.6       | 26.5           | 22.6            | 13.5            |
| Dacarbazine      | 200mg    | injection   | 6.8       | 6.3            | 8.8             | 6.1             |
| Dactinomycin     | 0.5mg    | injection   | 8.7       | 9.4            | 12.2            | 8.1             |
| Daunorubicin     | 20mg     | injection   | 19.3      | 10.9           | 10.7            | 9.4             |
| Docetaxel        | 20mg     | injection   | 5.2       | -              | -               | 15.8            |
| Docetaxel        | 80mg     | injection   | 17.5      | 21.8           | 27.6            | 38.5            |
| Doxorubicin      | 10mg     | injection   | 2.1       | 0.0            | 15.4            | 3.4             |
| Doxorubicin      | 50mg     | injection   | 5.4       | 4.3            | 7.5             | 7.9             |
| Gemcitabine      | 200mg    | injection   | 6.3       | 4.8            | 12.4            | 12.6            |
| Gemcitabine      | 1000mg   | injection   | 25.3      | 14.5           | 49.3            | 30.0            |
| Hydroxycarbamide | 500mg    | capsule     | 0.2       | 0.2            | 0.3             | 1.1             |
| Melphalan        | 2mg      | tablet      | 1.0       | 1.3            | 1.8             | 1.4             |
| Melphalan        | 50mg     | injection   | 45.5      | 0.0            | 0.0             | 64.0            |
| Mercaptopurine   | 50mg     | tablet      | 2.2       | 1.1            | 0.4             | 0.2             |
| Mercaptopurine   | 2.5mg    | tablet      | 0.1       | 0.1            | 0.1             | 0.1             |
| Oxaliplatin      | 50mg     | injection   | 28.9      | 12.8           | 34.0            | 16.1            |
| Oxaliplatin      | 100mg    | injection   | 74.8      | 15.4           | 46.5            | 25.7            |
| Paclitaxel       | 100mg    | injection   | 11.1      | 10.3           | 25.8            | 24.0            |
| Procarbazine     | 50mg     | tablet      | 1.4       | 0.8            | 1.1             | 0.8             |
| Tioguanine       | 40mg     | tab/cap     | 6.8       | -              | -               | -               |
| Vinblastine      | 10mg     | injection   | 2.6       | 8.1            | 9.0             | 7.7             |
| Vincristine      | 1mg      | injection   | 2.5       | 2.6            | 3.2             | 1.7             |
| Vincristine      | 2mg      | injection   | 4.1       | -              | -               | 8.1             |
| Vinorelbine      | 50mg     | injection   | 29.0      | 52.0           | 132.7           | 226.9           |
| Vinorelbine      | 10mg     | injection   | 22.0      | -              | -               | 65.0            |
| Imatinib         | 100mg    | tablet      | 0.7       | -              | -               | 1.1             |
| Imatinib         | 400mg    | tablet      | 25.2      | -              | 139.7           | 1.9             |
| Nilotinib        | 200mg    | capsule     | 1.2       | -              | 6.2             | 9.8             |

## **4.7 Affordability of Anticancer Drugs In Kenya**

According to WHO/HAI methodology applied in this study, a treatment is considered affordable if the course of treatment of a medicine is not more than the daily lowest government wage. To determine affordability, the lowest priced option out of the three sources was used to calculate the monthly expense. The lower strength of drugs that had multiple strengths was also selected because they were cheaper in most cases.

Treatment for cancer using most of the drugs surveyed is usually in cycles and rarely continuous. However, for the purpose of illustrating economic impact and comparison to other chronic conditions, 30 days were used to calculate the monthly expense of treatment.

In addition, due to their unique nature and application, most anticancer drugs have not been assigned conventional DDDs. Despite this, studies have been conducted with authors assigning DDDs based on standard treatment guidelines. For this study, this was applied and derived DDDs were used to calculate the monthly expenses.

According to the Wages Amendment 2022, Kenya's minimum monthly wage for unskilled personnel was set at KES 8109.90, which translates to a daily wage of KES 411. Based on the monthly expenses calculated, none of the surveyed drugs were affordable. Methotrexate 2.5mg at KES 420 was the closest to the daily wage. Highlighted in Table 4.16 are 10 of the least and most affordable drugs.



Table 4.16 Affordability of Selected Anticancer Drugs

| Medicine Name    | Strength | Monthly expense based on DDD (KES) | % of minimum wage |
|------------------|----------|------------------------------------|-------------------|
| Bendamustine     | 25mg     | 1065900                            | 13143.2           |
| Bevacizumab      | 100mg    | 1008000                            | 12429.3           |
| Cetuximab        | 100mg    | 456300                             | 5626.5            |
| Pembrolizumab    | 100mg    | 438662.4                           | 5409.0            |
| Rituximab        | 100mg    | 409500                             | 5049.4            |
| Bortezomib       | 3.5mg    | 310680                             | 3830.9            |
| Pertuzumab       | 420mg    | 287758.5                           | 3548.2            |
| Daunorubicin     | 10mg     | 249696                             | 3078.9            |
| Vinorelbine      | 10mg     | 166221                             | 2049.6            |
| Trastuzumab      | 150mg    | 92160                              | 1136.4            |
| Doxorubicin      | 10mg     | 6000                               | 74.0              |
| Mercaptopurine   | 50mg     | 6000                               | 74.0              |
| Carboplatin      | 150mg    | 5940                               | 73.2              |
| Cisplatin        | 10mg     | 4800                               | 59.2              |
| Asparaginase     | 10000iu  | 4326.3                             | 53.3              |
| Hydroxycarbamide | 500mg    | 2772                               | 34.2              |
| Vincristine      | 1mg      | 1800                               | 22.2              |
| Fluorouracil     | 250mg    | 1134                               | 14.0              |
| Calcium folinate | 15mg     | 525                                | 6.5               |
| Methotrexate     | 2.5mg    | 420                                | 5.2               |

#### 4.7.1 Affordability of Breast and Prostate Cancer Treatments

In addition to computing the monthly cost for each drug in the list, the cost for first line regimens for breast and prostate cancers were computed. These cancers were selected because they are the most prevalent in Kenya (53).

Three treatment regimens for breast and one for prostate cancers, were obtained from the Kenya Cancer Treatment Protocols and used to determine affordability of just the drugs used for one cycle of treatment. The parameters chosen for both patients were a weight of 70kg and body surface area of 1.7m<sup>2</sup>.

*Table 4.17 Cost of drugs used for one cycle treatment of breast and prostate cancers*

| <b>Provider</b>                                                                                         | <b>KNH<br/>(KES)</b> | <b>Cheapest<br/>Private Hosp<br/>(KES)</b> | <b>KEMSA<br/>(KES)</b> | <b>Cheapest<br/>Supplier<br/>(KES)</b> |
|---------------------------------------------------------------------------------------------------------|----------------------|--------------------------------------------|------------------------|----------------------------------------|
| <b>Cost</b>                                                                                             |                      |                                            |                        |                                        |
| 1 Cycle of chemotherapy(HER2 negative)<br>(Doxorubicin/Cyclophosphamide)                                | 2264.4               | 3982.08                                    | 1295.4                 | 1530                                   |
| 1 Cycle of chemotherapy(HER2 positive)<br>(Doxorubicin/Cyclophosphamide then<br>Paclitaxel/Trastuzumab) | 66735.6              | 45965                                      | 59560.6                | 32206.4                                |
| Cost of 1 cycle of biological<br>therapy(Trastuzumab)                                                   | 62363.6              | 35891                                      | 56633.2                | 28636.4                                |
| Cost of 1 cycle Docetaxel                                                                               | 4224                 | 3840                                       | -                      | 4000                                   |

As it can be observed in Table 4.17, none of the treatments were affordable and most would need at least a whole month's salary to cover the cost.

## 4.8 Linear regression analysis for identification of determinants of prices of anticancer drugs in Kenya

Linear regression was conducted to establish whether drug classes played a role in determining prices. Alkylating agents were used as the reference class and the other classes were compared against it. Table 4.18 shows the results of the regression. It can be observed that apart from antimetabolites, the rest of the classes were more expensive than alkylating agents with biologics having the highest difference.

Table 4.18 Linear regression analysis for Determinants of anticancer drug prices

| <b>Pharmacological Class</b>            | <b>Beta Coefficient</b> | <b>P</b> |
|-----------------------------------------|-------------------------|----------|
| <b>Alkylating Agents</b>                |                         |          |
| Antitumour                              | 2079.2 (-659.2,         | 0.131    |
| antibiotics                             | 4817.5)                 |          |
| Antimetabolites                         | -367.8 (-757.6,         | 0.064    |
|                                         | 21.9)                   |          |
| Plant Derived                           | 223.8 (-147.6,          | 0.230    |
| Agents                                  | 595.2 )                 |          |
| Biologics                               | 4875.4 (1096.4,         | 0.013    |
|                                         | 8654.4)                 |          |
| <b>Dosage form (Oral/Injectable)</b>    | -13200.8 (-             | 0.070    |
|                                         | 27507.8, 1106.1)        |          |
| <b>Originator/ Generics</b>             | -27593.8 (-             | 0.146    |
|                                         | 65145.8,                |          |
|                                         | 9958.3)                 |          |
| <b>Facility (KNH/ Private hospital)</b> | 361786.5 (-             | 0.297    |
|                                         | 325791.6,               |          |
|                                         | 1049365)                |          |
| <b>Multivariable Linear Regression</b>  |                         |          |
| Dosage Form                             | -17528.4 (-             | 0.012    |
|                                         | 31082.03, -             |          |
|                                         | 3974.773)               |          |
| Pharmacological                         | 4833.2 (                | 0.000    |
| Class                                   | 2265.051,               |          |
|                                         | 7401.348)               |          |

Linear regression was also conducted to establish whether prices differed depending on the dosage form. Injections were used as the reference group against tablets and capsules.

Table 4.18 shows the results where tablets and capsules were cheaper by KES 13200 (KES 1106, 27507).

Generics were also observed to have significantly lower prices to originator brands. On conducting linear regression on originator brands and lowest priced generic, it was observed that the generics were KES 27593 (KES 9958, 65145 ) cheaper than originator brands. Therefore, presence of generics for different drugs determine prices of anticancer drugs within the market.

It was also observed that second tier suppliers stocked generics over originator brands. Therefore, depending on where drugs are sourced from tier 1 or 2 suppliers, prices may differ significantly.

Finally, it is often thought that private facilities are more expensive than public facilities. From price comparisons done in the earlier section between KNH and one private hospital, it was observed that either hospital had half of the drugs cheaper than the other. This dispels the common belief that private hospitals are usually more expensive than public hospitals. In addition, linear regression conducted for KNH and Private hospital 3 showed that KNH was more expensive by KES 361786 (KES 325792, 1049365) when it came to the drugs that were common between the two hospitals.

## 5 DISCUSSION

The main objective of this study was to assess the availability, price and affordability of anticancer drugs in Nairobi, Kenya. This study found that availability of drugs across all providers was below WHO's set benchmark and all drugs assessed were unaffordable according to WHO/HAI criteria. In addition, prices varied across the different cancer care providers surveyed.

### 5.1 Availability of Anticancer Drugs

The list of drugs adopted contained anticancer drugs on the KEML, which matched WHO Essential Medicines List for anticancer drugs. Most LMICs rely on WHO's EML to prepare their own lists and this guides procurement and acquisition of medicines for the public sector (54). WHO sets the baseline standard for availability of essential medicines required to treat NCDs, which includes cancer at 80% (55). A number of anticancer drugs were considered essential medicines so as to increase their accessibility especially to LMICs who rely on WHO to set the guidelines due to limited budgetary allocations (56). In this study, none of the drugs surveyed in their different formulations achieved this standard of 80%. The closest drugs to this mark included; carboplatin 450mg (78.9%), capecitabine 500mg (73.7%), docetaxel 80mg (73.7%), doxorubicin 50mg (73.7%), rituximab 500mg (73.7%) and trastuzumab 440mg (73.7%). The drugs with the least availability at 5.3% include; vincristine 2mg, daunorubicin 10mg, chlorambucil 5mg, melphalan 50mg and thalidomide 50mg. Arsenic trioxide and tioguanine were completely unavailable from all surveyed providers. A study conducted in Ethiopia had some similarities in a number of drugs that had low availability and found a significant number of drugs unavailable such as; bendamustine, bleomycin, chlorambucil, fludarabine and paclitaxel (57). The low availability can be attributed to inadequate investment in cancer care and insufficient incentives to stock anticancer drugs (51). In addition, the surveyed providers may not necessarily be following the Kenya Essential Medicines List leading to

differences in drug catalogues and as a result inconsistency in drugs available. To remedy this, the government can create policy where treatment of anticancer patients results in reimbursement by the national insurance scheme as it currently does for maternity and renal patients. This will result in more hospitals stocking anticancer drugs and consequently, increased availability. A testament to this approach was seen in China when 18 anticancer drugs were added to the National Reimbursement Drug List Negotiation and this resulted to increased availability (58).

All classes of drugs were available evenly across all the anticancer providers with plant derived agents (41.9%) slightly edging ahead of alkylating agents, and antimetabolites (39.6%) and finally molecular targeted agents (37.6%). This is consistent with a study done in Punjab, Pakistan that found non-biologic agents were more available than biologic agents (59). This can be attributed to the higher prices for biologics, which makes them unaffordable for most. In addition, prolonged patent periods and data exclusivity also limit the availability of biologic drugs (59). However, biologics are increasingly being applied in treatment of cancer due to their targeted action and ability to personalize therapy (60). It is therefore important to increase their availability in the country. This can be done by opting for biosimilars where possible and informing prescribers of their presence in the market and consequently reduce overreliance on originator biologics (59). In addition, the government can negotiate with the multinational companies manufacturing the originator brands and discuss subsidized prices to the hospitals resulting in increased availability. The Chinese government negotiated the prices of Gefitinib and Icotinib which are biological agents and this resulted in increased availability at the hospitals (61).

Lowest priced generics (LPGs) (37%) were more available than originator brands (OBs) (3.6%) across all providers. A similar situation was observed in Ethiopia with LPGs accounting for 20.21% and OBs at 1.89% (57). This can be attributed to the significant price differences which make LPGs more accessible for LMICs (62). However, a study conducted in Pakistan found that OBs (52.5%) were more available than LPGs (28.1%).

The authors attributed this to multinational pharmaceutical companies, which adopt strategies such as marketing, and patent rights to ensure their products are the preferred choice (62). OBs are important where LPGs for the molecules are not yet available due to patents. As a result, government needs to negotiate with multinational pharmaceutical companies and avail them to hospitals for ease of access.

Hospitals (50.4%) had the highest availability of the surveyed drugs, followed by suppliers (39%) and finally supply agencies (29%). The high availability in the hospitals can be attributed to KNH (45.7%), which is the country's largest referral hospital and to the private sector hospitals (34.8%, 58.7% and 64.1%). This trend was also seen in a study done in New Delhi, where availability of antineoplastic agents for treating childhood cancers was 43% in public hospital pharmacies and 71% in private hospital pharmacies (63). The availability of anticancer drugs was greater in the private sector compared to the public sector. Similarly, studies conducted in Pakistan and Rwanda, revealed that availability of anticancer medicines was higher in the private sector compared to government facilities (62) (64). In contrast, a study done in Ethiopia found that the public sector had a greater contribution to the mean availability of cytotoxic medications (57). The differences in both sectors can be attributed to the difference in purchasing power of government and private institutions. The availability in the surveyed providers however, did not reach the 80% benchmark set by WHO. The low availability in suppliers can be attributed to low demand and as a result, fear of losses due to expiries if drugs are stocked but not sold (57). The government should put in place policies such as tax incentives to private and public sector providers who offer cancer care and in so doing stimulate availability of anticancer medicines.

## **5.2 Prices of Anticancer Drugs**

The difference in cost between various brands of the same drug can vary from two-fold to 100 fold due to various reasons such as drugs being under patent protection and sellers dictating the price of the different products (65). In addition, the geographic location may

act as a factor in price variation because different prices exist for the same medicine in different health systems (66). This study found significant price variations for the different drugs surveyed.

The highest price variations were observed for antimetabolite agents such as Mercaptopurine 50mg (1471.4%) and Cytarabine 1000mg (1400%). This was in contrast to a study conducted in Nepal that found minimal price variation for this class of drugs (67). For molecular targeted agents, Imatinib 400mg had the highest price variation of 1454.4%. However, contrary to this, a study in India which found Imatinib 400mg with a 0% price ratio and Imatinib 100mg with a the highest price variation of targeted agents at 5.6% (65). For natural agents, this study found docetaxel 20mg to have the highest price variation at 1405.3%, which again contrasts to the Indian study that found paclitaxel to have the lowest (65). However, Shrestha et al, found that natural products had the highest price variation in Nepal (67). The price variations in this study were quite vast (1471.4%) compared to the Indian study (146.9%) and the Nepalese study (590.3%). This can be attributed to the numerous anticancer drug manufacturers in the Indian pharmaceutical sector. This implies that depending on where the drugs are sourced from, patients access the drugs at substantially different prices and as a result, some are disadvantaged. There is need to standardize prices and for government to put in place price checks for anticancer drugs to ensure equitable access to the patients. Additionally, policy on transparency of drug pricing should be introduced in the pharmaceutical sector to prevent commercialization of cancer treatment. Global practice has revealed that transparent pricing and better regulation play a great role in preventing high prices and variations drugs (67).

A comparison between KNH and the private hospital with the lowest prices found that half of the drugs were cheaper from the private hospital. Similarly, a Nepalese study found that prices in two private hospitals were either equal to or lower than prices set in government hospitals (67). However, this was in contrast to the Ethiopian study which found, higher private sector prices compared to public sector (57). This can be explained



by the differences in procurement methods used between KNH and the private sector. KNH being a public institution is required by law to follow public procurement procedures stipulated in the constitution whereas, private institutions can buy from the cheapest suppliers and in some cases even import for themselves. However, public procurement in Kenya is riddled with corruption and as a result prices of commodities sometimes end up being inflated and this is passed on to the patient (68). In order to curb this, procurement of drugs in public institutions should be transparent and drug prices subjected to audits to ensure access at the patient level.

Pricing of essential medicines is not regulated in Kenya leading to large mark-ups along the distribution chain, which contribute to the final price (69). The accepted mark-up in most retail pharmacies, which serve the population, is 30-40%. This study found significant mark-ups both in public and private hospitals. This can be attributed to the lack of pricing controls in the country and limited number of anticancer care providers. In India, the National Pharmaceutical Pricing Authority capped the trade margins for 42 anticancer drugs at 30% and this saw a reduction of up to 90% in the maximum retail prices (70). In Jiangsu Province, a “zero-markup” in public institutions was adopted and this led to reduced revenues for the hospitals which translated to poor stocking of essential medicines (51). The Kenyan government should learn from both scenarios and implement a transparent drug pricing system and efficient regulatory mechanisms, which will ensure adequate revenues that, will sustain procurement of drugs.

International reference pricing is important because it provides a benchmark for most countries to restrict drug costs (54). A comparison of median unit prices from the three main providers showed that most were comparable to international reference prices apart from a few drugs. In Mexico, anticancer medicines overall were found to be 20% lower than the MSH reference price (55). These findings indicate that Kenya’s drug prices are comparable to international prices but still not accessible to most. Additional efforts should be put in place to lower the prices below the reference prices, as is the case in Mexico.

### **5.3 Affordability of Anticancer Drugs**

This study found that all drugs were unaffordable and patients needed to pay more than the daily wage to get the medicines. This was in line with studies conducted in neighbouring countries Tanzania, Ethiopia and Rwanda that found anticancer drugs were unaffordable (57) (64) (71). The picture for LMICs outside Africa was also consistent with these findings. A study conducted in Mexico found essential cancer medicines still unaffordable (55). This worsens poverty in LMICs and the patient's outcome.

The lowest prices for the drugs surveyed were selected to determine affordability and despite this, treatments were still found to be unaffordable. Similarly, studies conducted in LMICs have shown that even LPGs for essential medicines are still unaffordable (72). Furthermore, treatment regimens contain more than one drug, which increases the total cost to the patient. In contrast, Sarwar et al, found a few LPGs for drugs such as cytarabine, fluorouracil and methotrexate, which were affordable to both low and high-income populations in Pakistan (62). Affordability is directly affected by drug prices therefore, it is important to put in place aggressive pricing policies to curb unaffordability (55). Studies have identified access to LPGs as a way of increasing affordability (51). Government should therefore, implement policies and regulations that make approvals of generic anticancer medicines faster and improve their access to the market (51).

The most unaffordable class was molecular targeted agents, which made up seven out of ten of the least affordable drugs. This was mirrored in a study conducted in Punjab that found biologics to be more unaffordable than non-biologics despite them being OBs or LPGs (59). As earlier mentioned, this can be attributed to higher prices and prolonged patent durations, which limit development of biosimilars.

Breast and prostate cancers are the most prevalent in Kenya for women and men respectively (53). The two were assessed for affordability of their treatment using drugs

that were surveyed and it was found that a male patient would need at least 10 days' wage and a female patient would need 3 days' wage, to just pay for the drugs for one treatment cycle. Similarly, a study conducted in New Delhi found 244 days' wages were needed to pay for the treatment of acute lymphoblastic leukemia in children (63). These high costs put patients at a higher risk of poverty once diagnosed with cancer. The government should ensure that the national insurance scheme is accessible to the low-income populations and treatment of cancer is covered. This will in turn cushion them from depleting their incomes when accessing treatment.

#### **5.4 Determinants of Anticancer Drug Prices**

This study sought to determine the factors that influence prices and eventual affordability of anticancer drugs. The factors in focus were drug related factors, supply chain factors and facility related factors.

A number of drug related factors such as class and presence of generics affect the prices of anticancer drugs. From the linear regression that was conducted, alkylating agents and antimetabolites were the cheapest classes with naturally sourced agents and biologics being more expensive. Biologics were the most expensive of all classes. However, the presence of biosimilars in the market reduced the prices significantly. For example, the originator brand for bevacizumab 400mg was priced at KES 180,000 at one supplier, while a biosimilar was priced at KES 52,000 from a different supplier. Despite this, the prices still deem most drugs inaccessible. As earlier mentioned, negotiations with drug manufacturers and stringent price regulations can be applied to reduce the prices.

The higher cost for antitumour antibiotics and plant derived agents can be attributed to the cost of obtaining drugs from natural sources, especially plants because the chemicals accumulate in the plant in very small quantities over long durations. Their chemical structures also render the total synthesis impractical. Therefore, manufacturers result to semi-synthesis using precursor molecules, which still has a high cost of production (36).

In response to this, methods of production of these agents using microbes such as bacteria and yeast are being explored in order to reduce the high cost of production (36). This can then be passed on to the patients in reduced drug prices.

The regression analysis also confirmed that generics are indeed cheaper than originator brands. An interesting observation from this study was that tier one suppliers mostly stocked originator brands for anticancer drugs whose prices are significantly higher while tier two suppliers mostly stocked generics. A few suppliers stocked both options for the same drugs. In addition, private hospitals were found to have a number of drugs cheaper than the public sector and even suppliers. This was as a result of directly importing their own drugs. Therefore, the approach of sourcing drugs can determine the prices and eventual affordability. The supply chain of drugs in Kenya is not that well regulated leading to the different sourced and priced drugs. This lack of regulation has led to very low priced drugs due to increased competition by the different players. For anticancer drugs however, the providers are still limited and the prices are still high. Pooled procurement for cancer care providers can be explored as a way of obtaining anticancer drugs so as to pass on the reduced cost to patients.

Finally, regression analysis conducted to compare the prices between KNH and Private hospital 3 showed that KNH was more expensive. This can be attributed to procurement practices applied in KNH as a result of being a government institution and the ability of some of the private hospitals to import drugs which translates to lower consumer prices. Therefore, public facilities may not necessarily offer anticancer drugs at cheaper prices.

## 5.5 Study Strengths and Limitations

This study surveyed the main sources of anticancer drugs in Nairobi, where most of the treatment of cancer is done within the country. It employed a methodology of mystery shopping that reduced information bias and acquired the real time data.

However, the study was conducted only in Nairobi which may not indicate the whole situation in the country which might be worse than these findings.

At the onset, this study was prone to non-response bias therefore, to overcome this; a waiver for informed consent was sought and obtained from KNH-UON ERC. Mystery shopping technique was employed to try to avoid the non-response bias. However, the assistants still encountered difficulties especially when it came to private hospitals. In one of the hospitals, one of the assistants was advised to refer their patient to the hospital directly instead of inquiring for drug prices. In another hospital, the assistant was denied access to the oncology pharmacy. As a result, the total number of hospitals surveyed was four, three of them private. This further points to the lack of transparency that has plagued the healthcare system especially when it comes to private facilities.

The MSH Guidelines for International Reference Pricing were outdated and hence did not include a number of the drugs surveyed during this study. The value of the dollar has also changed since 2015 therefore, comparison with the reference price may not have been reliable. Despite that, the reference list still provided insights into the prevailing situation.

## **6 CONCLUSION AND RECOMMENDATIONS.**

### **6.1 Conclusion**

The availability of anticancer drugs was highest in the hospitals, followed by suppliers and finally supply agencies. The surveyed prices of anticancer drugs compared favourably with international reference prices. Prices of anticancer drugs varied across the different providers and the mode of procurement was key to the end user price. In addition, prices of anticancer drugs were determined by pharmacological class, dosage form, type of provider and whether the drug was generic or originator brand. The findings also showed that none of the drugs surveyed were affordable according to WHO/HAI methodology.

### **6.2 Recommendations for Policy and Practice**

Supply agencies should improve their levels of stocking anticancer drugs, which will translate to increased availability to public, and mission hospitals that serve a greater population. Transparency of drug pricing methodology across all sectors should also be enforced to limit exploitation of patients. Furthermore, drug mark-ups in the health system should be actively monitored to limit excessive profits and enhance affordability.

### **6.3 Recommendations for Future Research**

Similar research should be conducted in hospitals outside Nairobi that are offering cancer care so as to provide a clearer picture of the economic implication of cancer treatment in other parts of the country. In addition, conducting a similar research outside Nairobi can establish availability of drugs within these parts and may be able to explain the overreliance on Nairobi as a treatment hub for cancer.

## REFERENCES

1. Bray F, Ferlay J, Soerjomataram I, Siegel RL, Torre LA, Jemal A. Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin*. 2018 Nov;68(6):394–424.
2. Cancer [Internet]. [cited 2020 Dec 28]. Available from: <https://www.who.int/news-room/fact-sheets/detail/cancer>
3. Thun MJ, Linet MS, Cerhan JR, Haiman C, Schottenfeld D, editors. *Schottenfeld and Fraumeni cancer epidemiology and prevention*. Fourth edition. New York, NY: Oxford University Press; 2018. 1308 p.
4. Atieno OM, Oponga S, Martin A, Kurdi A, Godman B. Pilot study assessing the direct medical cost of treating patients with cancer in Kenya; findings and implications for the future. *J Med Econ*. 2018 Sep 2;21(9):878–87.
5. Jemal A, Torre L, Soerjomataram I, Bray F, editors. *The cancer atlas*. Third edition. Atlanta: American Cancer Society; 2019.
6. Makau-Barasa LK, Greene S, Othieno-Abinya NA, Wheeler SB, Skinner A, Bennett AV. A review of Kenya’s cancer policies to improve access to cancer testing and treatment in the country. *Health Res Policy Syst*. 2020 Dec;18(1):2.
7. Stewart BW, Wild C, International Agency for Research on Cancer, World Health Organization. *World cancer report 2014*. 2014.
8. Cancer and its status in Kenya [Internet]. [cited 2021 Jan 2]. Available from: <https://kippra.or.ke/index.php/resource-centre/blogs/155-cancer-and-it-s-status-in-kenya>
9. Chalkidou K, Marquez P, Dhillon PK, Teerawattananon Y, Anothaisintawee T, Gadelha CAG, et al. Evidence-informed frameworks for cost-effective cancer care and prevention in low, middle, and high-income countries. *Lancet Oncol*. 2014 Mar;15(3):e119–31.
10. Siddiqui M, Rajkumar SV. The High Cost of Cancer Drugs and What We Can Do About It. *Mayo Clin Proc*. 2012 Oct;87(10):935–43.
11. Kantarjian H, Steensma D, Rius Sanjuan J, Elshaug A, Light D. High Cancer Drug Prices in the United States: Reasons and Proposed Solutions. *J Oncol Pract*. 2014 Jul;10(4):e208–11.
12. Cohen D. Cancer drugs: high price, uncertain value. *BMJ*. 2017 Oct 4;j4543.
13. Dolgin E. The price trajectory of oncology drugs is unsustainable — but fixes are in the works. *Nature*. 555:4.
14. Prasad V, De Jesús K, Mailankody S. The high price of anticancer drugs: origins, implications, barriers, solutions. *Nat Rev Clin Oncol*. 2017 Jun;14(6):381–90.

15. Gilbar PJ, Chambers CR. How can we ensure value for money from expenditure on injectable cancer drugs? *J Oncol Pharm Pract*. 2018 Sep;24(6):473–6.
16. Howard DH, Bach PB, Berndt ER, Conti RM. Pricing in the Market for Anticancer Drugs. *J Econ Perspect*. 2015 Feb 1;29(1):139–62.
17. Saluja R, Arciero VS, Cheng S, McDonald E, Wong WWL, Cheung MC, et al. Examining Trends in Cost and Clinical Benefit of Novel Anticancer Drugs Over Time. *J Oncol Pract*. 2018 May;14(5):e280–94.
18. DeMartino PC, Miljković MD, Prasad V. Potential Cost Implications for All US Food and Drug Administration Oncology Drug Approvals in 2018. *JAMA Intern Med*. 2021 Feb 1;181(2):162–7.
19. Adedeji OA, editor. *Cancer in Sub-Saharan Africa* [Internet]. Cham: Springer International Publishing; 2017 [cited 2021 Jan 8]. Available from: <http://link.springer.com/10.1007/978-3-319-52554-9>
20. Kantarjian H, Rajkumar SV. Why Are Cancer Drugs So Expensive in the United States, and What Are the Solutions? *Mayo Clin Proc*. 2015 Apr;90(4):500–4.
21. Kajilwa G. Health. [cited 2021 Jan 8]. NHIF to foot bills of cancer patients in full - Standard Health. Available from: <https://www.standardmedia.co.ke/health/article/2001312856/nhif-to-foot-bills-of-cancer-patients-in-full>
21. World Health Organization. (2019). Global spending on health: a world in transition. World Health Organization. <https://apps.who.int/iris/handle/10665/330357>. License: CC BY-NC-SA 3.0 IGO
23. Lu Y, Hernandez P, Abegunde D, Edejer T. The World Medicines Situation 2011. :34.
24. Hartman M, Martin AB, Espinosa N, Catlin A, The National Health Expenditure Acc. National Health Care Spending In 2016: Spending And Enrollment Growth Slow After Initial Coverage Expansions. *Health Aff (Millwood)*. 2018 Jan;37(1):150–60.
25. Xi X, Li W, Li J, Zhu X, Fu C, Wei X, et al. A survey of the availability, prices and affordability of essential medicines in Jiangsu Province, China. *BMC Health Serv Res*. 2015 Dec;15(1):345.
26. World Health Organization, World Health Organization, Department of Essential Medicines and Health Products. WHO guideline on country pharmaceutical pricing policies. [Internet]. 2015 [cited 2020 Dec 26]. Available from: [http://apps.who.int/iris/bitstream/10665/153920/1/9789241549035\\_eng.pdf?ua=1](http://apps.who.int/iris/bitstream/10665/153920/1/9789241549035_eng.pdf?ua=1)
27. WHO [Internet]. World Health Organization; [cited 2021 Apr 25]. WHO | The Abuja Declaration: Ten Years On. Available from: [http://www.who.int/healthsystems/publications/abuja\\_declaration/en/](http://www.who.int/healthsystems/publications/abuja_declaration/en/)



28. David N, Wanjala P. A Case for Increasing Public Investments in Health. :8.
28. Economist Intelligence Unit (EIU). Breakaway: The global burden of cancer -challenges and opportunities. A report from the Economist Intelligence Unit. In: The Economist [Internet]. 2009 \_ [cited 25 Apr 2021]  
[http://graphics.eiu.com/upload/eb/EIU\\_LIVESTRONG\\_Global\\_Cancer\\_Burden.pdf](http://graphics.eiu.com/upload/eb/EIU_LIVESTRONG_Global_Cancer_Burden.pdf)
30. The Global Economic Cost of Cancer [Internet]. [cited 2021 Apr 25]. Available from: <https://www.phrma.org/Report/The-Global-Economic-Cost-of-Cancer>
31. de Oliveira C, Weir S, Rangrej J, Krahn MD, Mittmann N, Hoch JS, et al. The economic burden of cancer care in Canada: a population-based cost study. CMAJ Open. 2018 Jan 4;6(1):E1–10.
32. Earp Siqueira A de S, Goncalves JG, Xavier Mendonca PE, Elias Merhy E, Poirot Land MG. Economic Impact Analysis of Cancer in the Health System of Brazil: Model Based in Public Database. Health Sci J [Internet]. 2017 [cited 2020 Jun 27];11(4). Available from: <http://www.hsj.gr/medicine/economic-impact-analysis-of-cancer-in-the-health-system-of-brazil-model-based-in-public-database.php?aid=19776>
32. World Health Organization. (2018). Technical report: pricing of cancer medicines and its impacts: a comprehensive technical report for the World Health Assembly Resolution 70.12: operative paragraph 2.9 on pricing approaches and their impacts on availability and affordability of medicines for the prevention and treatment of cancer. World Health Organization. <https://apps.who.int/iris/handle/10665/277190>. License: CC BY-NC-SA 3.0 IGO
34. Truong J, Chan KKW, Mai H, Chambers A, Sabharwal M, Trudeau ME, et al. The impact of pricing strategy on the costs of oral anti-cancer drugs. Cancer Med. 2019 May 27;8(8):3770–81.
35. Light DW., Kantarjian H. Market spiral pricing of cancer drugs: Market Spiral Pricing of Cancer Drugs. Cancer. 2013 Nov 15;119(22):3900–2.
36. Courdavault V, O'Connor SE, Oudin A, Besseau S, Papon N. Towards the Microbial Production of Plant-Derived Anticancer Drugs. Trends Cancer. 2020 Jun;6(6):444–8.
35. European Society for Medical Oncology (ESMO). New Studies Question Whether Novel Anti-Cancer Drugs Are Worth Their Extra Cost [ESMO 2019 Press Release] [Internet]. [cited 2020 Aug 20]. Available from: <https://www.esmo.org/newsroom/press-office/esmo-congress-anticancer-drugs-cost-marino-vokinger>
38. Osman OM. Meet the Kenyans too poor to afford cancer treatment [Internet]. [cited 2021 Apr 25]. Available from: <https://www.aljazeera.com/features/2016/2/4/meet-the-kenyans-too-poor-to-afford-cancer-treatment>

39. Cameron A, Ewen M, Ross-Degnan D, Ball D, Laing R. Medicine prices, availability, and affordability in 36 developing and middle-income countries: a secondary analysis. *The Lancet*. 2009 Jan;373(9659):240–9.
40. Jiang M, Yang S, Yan K, Liu J, Zhao J, Fang Y. Measuring Access to Medicines: A Survey of Prices, Availability and Affordability in Shaanxi Province of China. Beck EJ, editor. *PLoS ONE*. 2013 Aug 1;8(8):e70836.
41. Iyengar S, van den Ham R, Suleman F. Medicine Prices in Africa. In: *Medicine Price Surveys, Analyses and Comparisons* [Internet]. Elsevier; 2019 [cited 2020 Dec 27]. p. 85–111. Available from: <https://linkinghub.elsevier.com/retrieve/pii/B9780128131664000073>
42. Babar ZU din, Hussain R, Hasan SS. Medicine Prices in Asia–Pacific Countries. In: *Medicine Price Surveys, Analyses and Comparisons* [Internet]. Elsevier; 2019 [cited 2020 Dec 27]. p. 45–60. Available from: <https://linkinghub.elsevier.com/retrieve/pii/B978012813166400005X>
43. Health Action International [Internet]. [cited 2021 Apr 25]. Survey & Related Reports. Available from: <https://haiweb.org/survey-related-reports/>
44. Health Action International [Internet]. [cited 2021 Apr 25]. Prices and Availability of Locally Produced and Imported Medicines in Kenya. Available from: <https://haiweb.org/publication/prices-and-availability-of-locally-produced-and-imported-medicines-in-kenya/>
45. WHO. WHO. World Health Organization; [cited 2021 Apr 25]. WHO | Measuring medicine prices, availability, affordability and price components. Available from: [http://www.who.int/medicines/areas/access/medicines\\_prices08/en/](http://www.who.int/medicines/areas/access/medicines_prices08/en/)
46. Wambalaba FW, Son B, Wambalaba AE, Nyong’o D, Nyong’o A. Prevalence and Capacity of Cancer Diagnostics and Treatment: A Demand and Supply Survey of Health-Care Facilities in Kenya. *Cancer Control*. 2019 Jan 1;26(1):107327481988693.
47. Ministry Of Health. Kenya Essential Medicines List 2019 – MINISTRY OF HEALTH [Internet]. [cited 2021 Apr 25]. Available from: <https://www.health.go.ke/kenya-essential-medicines-list-2019/>
48. Ministry Of Health. Kenya National Cancer Treatment Protocols 2019 [Internet]. Ministry of Health; 2019 [cited 2021 Apr 25]. Available from: <http://repository.kippra.or.ke/handle/123456789/1761>
49. Mercurio N, Flesta J. Designing a ‘Smarter’ Mystery Shopping Program. :10.
50. Mináriková D, Fazekas T, Minárik P, Jurišová E. Assessment of patient counselling on the common cold treatment at Slovak community pharmacies using mystery shopping. *Saudi Pharm J*. 2019 May;27(4):574–83.

51. Zhu Y, Wang Y, Sun X, Li X. Availability, Price and Affordability of Anticancer Medicines: Evidence from Two Cross-Sectional Surveys in the Jiangsu Province, China. *Int J Environ Res Public Health* [Internet]. 2019 Oct [cited 2021 Jan 12];16(19). Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6801951/>
52. Frye JE. 2015 International Medical Products Price Guide. :443.
53. Rooprai PJ HOPE CRAIG, CHANDRAKANTH ARE, Gurpreet. Cancer on the Global Stage: Incidence and Cancer-Related Mortality in Kenya - The ASCO Post [Internet]. [cited 2023 Feb 15]. Available from: <https://ascopost.com/issues/february-25-2021/cancer-on-the-global-stage-incidence-and-cancer-related-mortality-in-kenya/>
54. Ocran Mattila P, Ahmad R, Hasan SS, Babar ZUD. Availability, Affordability, Access, and Pricing of Anti-cancer Medicines in Low- and Middle-Income Countries: A Systematic Review of Literature. *Front Public Health* [Internet]. 2021 [cited 2023 Feb 15];9. Available from: <https://www.frontiersin.org/articles/10.3389/fpubh.2021.628744>
55. Moye-Holz D, Ewen M, Dreser A, Bautista-Arredondo S, Soria-Saucedo R, van Dijk JP, et al. Availability, prices, and affordability of selected essential cancer medicines in a middle-income country – the case of Mexico. *BMC Health Serv Res*. 2020 May 14;20(1):424.
56. Jenei K, Aziz Z, Booth C, Cappello B, Ceppi F, de Vries EGE, et al. Cancer medicines on the WHO Model List of Essential Medicines: processes, challenges, and a way forward. *Lancet Glob Health*. 2022 Dec;10(12):e1860–6.
57. Alemu BA, Hailemariam FH. Price, Availability and Affordability of Anti-Cancer Medicines in Addis Ababa, Ethiopia. *Risk Manag Healthc Policy*. 2022 Dec;Volume 15:2421–33.
58. Zhu H, Zhu J, Zhou Y, Shan L, Li C, Cui Y, et al. Impact of the National Reimbursement Drug List Negotiation Policy on Accessibility of Anticancer Drugs in China: An Interrupted Time Series Study. *Front Public Health* [Internet]. 2022 [cited 2023 Mar 31];10. Available from: <https://www.frontiersin.org/articles/10.3389/fpubh.2022.921093>
59. Saqib A, Iftikhar S, Sarwar MR. Availability and affordability of biologic versus non-biologic anticancer medicines: a cross-sectional study in Punjab, Pakistan. *BMJ Open*. 2018 Jun 1;8(6):e019015.
60. Shinde VA, Patil R, Bhandari P, Borkar P, Yadav S, Shinde VA, et al. A Pharmacoeconomic Outlook of the Biological Drugs Marketed in India: A Cost Variance Analysis. *Cureus* [Internet]. 2023 Jan 18 [cited 2023 Mar 30];15(1). Available from: <https://www.cureus.com/articles/122668-a-pharmacoeconomic-outlook-of-the-biological-drugs-marketed-in-india-a-cost-variance-analysis>
61. Huang C, Ung COL, Wushouer H, Bai L, Li X, Guan X, et al. Trends of Negotiated Targeted Anticancer Medicines Use in China: An Interrupted Time Series Analysis. *Int J Health Policy Manag*. 2021 Jun 9;1.

62. Sarwar MR, Iftikhar S, Saqib A. Availability of anticancer medicines in public and private sectors, and their affordability by low, middle and high-income class patients in Pakistan. *BMC Cancer*. 2018 Dec;18(1):14.
63. Faruqui N, Martiniuk A, Sharma A, Sharma C, Rathore B, Arora RS, et al. Evaluating access to essential medicines for treating childhood cancers: a medicines availability, price and affordability study in New Delhi, India. *BMJ Glob Health*. 2019 Apr;4(2):e001379.
64. Rurangwa C, Ndayisenga J, Sezirahiga J, Nyirimigabo E. Availability and Affordability of Anticancer Medicines at Cancer Treating Hospitals in Rwanda [Internet]. In Review; 2022 Nov [cited 2023 Feb 15]. Available from: <https://www.researchsquare.com/article/rs-2238542/v1>
65. Kolasani BP. Variation of Cost among Anti-cancer Drugs Available in Indian Market. *J Clin Diagn Res* [Internet]. 2016 [cited 2023 Feb 8]; Available from: [http://jcdmr.net/article\\_fulltext.asp?issn=0973-709x&year=2016&volume=10&issue=11&page=FC17&issn=0973-709x&id=8918](http://jcdmr.net/article_fulltext.asp?issn=0973-709x&year=2016&volume=10&issue=11&page=FC17&issn=0973-709x&id=8918)
66. Cuomo RE, Seidman RL, Mackey TK. Country and regional variations in purchase prices for essential cancer medications. *BMC Cancer*. 2017 Aug 24;17(1):566.
67. Shrestha S, Poudel RS, Kc B, Poudel BK, Sapkota B, Sharma S, et al. Price variation among different brands of anticancer medicines available in hospital pharmacies of Nepal. *J Pharm Policy Pract*. 2020 Dec;13(1):6.
68. Mutangili SK. Corruption in Public Procurement in Kenya: Causes, Consequences, Challenges and Cures. *Supply Chain*. 3(1).
69. Wamae W, Kungu JK, Clark N, Mackintosh M. Spotlight on pharmaceutical pricing regulation in Kenya: how much does it really contribute to access?
70. Cap on trade margin of 42 non-scheduled cancer drugs by NPPA led to up to 90 pc reduction in MRP: Mansukh Mandaviya. *The Economic Times* [Internet]. 2021 Aug 10 [cited 2023 Feb 14]; Available from: <https://economictimes.indiatimes.com/industry/healthcare/biotech/pharmaceuticals/cap-on-trade-margin-of-42-non-scheduled-cancer-drugs-by-nppa-led-to-up-to-90-pc-reduction-in-mrp-mansukh-mandaviya/articleshow/85208815.cms>
71. Yohana E, Kamuhabwa A, Mujinja P. Availability and affordability of anticancer medicines at the Ocean Road Cancer Institute in Dar es Salaam, Tanzania. *East Afr J Public Health*. 2011 Mar;8(1):52–7.
72. Eden T, Burns E, Freccero P, Renner L, Paintsil V, Dolendo M, et al. Are essential medicines available, reliable and affordable in low-middle income countries? *J Cancer Policy*. 2019 Mar;19:100180.

## 7 APPENDICES.

### 7.1 APPENDIX A: DATA COLLECTION FORM.

#### Data Collection Form.

Form Number: \_\_\_\_\_

Date: \_\_\_\_\_

#### 1. Facility Information.

Facility/Institution name: \_\_\_\_\_

Location: \_\_\_\_\_

Type of facility:

- i. Private Hospital ☐
- ii. Public Hospital ☐
- iii. Wholesale Supplier ☐
- iv. Retail Supplier ☐

#### 2. Medicine Information.

- i. Are these drugs available in your facility and at what price for each unit dose?

| Medicine Number | Medicine Name    | Strength | Dosage Form | Present in KEML | Unit Price | Originator Brand or Generic |
|-----------------|------------------|----------|-------------|-----------------|------------|-----------------------------|
| 1               | Arsenic trioxide |          |             |                 |            |                             |
| 2               | Asparaginase     |          |             |                 |            |                             |
| 3               | Bendamustine     |          |             |                 |            |                             |
| 4               | Bleomycin        |          |             |                 |            |                             |
| 5               | Calcium folinate |          |             |                 |            |                             |
| 6               | Capecitabine     |          |             |                 |            |                             |
| 7               | Carboplatin      |          |             |                 |            |                             |
| 8               | Chlorambucil     |          |             |                 |            |                             |
| 9               | Cisplatin        |          |             |                 |            |                             |
| 10              | Cyclophosphamide |          |             |                 |            |                             |
| 11              | Cytarabine       |          |             |                 |            |                             |
| 12              | Dacarbazine      |          |             |                 |            |                             |

|    |                         |  |  |  |  |  |
|----|-------------------------|--|--|--|--|--|
| 13 | Dactinomycin            |  |  |  |  |  |
| 14 | Daunorubicin            |  |  |  |  |  |
| 15 | Docetaxel               |  |  |  |  |  |
| 16 | Doxorubicin             |  |  |  |  |  |
| 17 | Etoposide               |  |  |  |  |  |
| 18 | Fluorouracil            |  |  |  |  |  |
| 19 | Gemcitabine             |  |  |  |  |  |
| 20 | Hydroxycarbamide        |  |  |  |  |  |
| 21 | Ifosfamide + mesna      |  |  |  |  |  |
| 22 | Imatinib                |  |  |  |  |  |
| 23 | Irinotecan              |  |  |  |  |  |
| 24 | Liposomal Doxorubicin   |  |  |  |  |  |
| 25 | Melphalan               |  |  |  |  |  |
| 26 | Mercaptopurine          |  |  |  |  |  |
| 27 | Methotrexate            |  |  |  |  |  |
| 28 | Oxaliplatin             |  |  |  |  |  |
| 29 | Paclitaxel              |  |  |  |  |  |
| 30 | Procarbazine            |  |  |  |  |  |
| 31 | Tioguanine              |  |  |  |  |  |
| 32 | Vinblastine             |  |  |  |  |  |
| 33 | Vincristine             |  |  |  |  |  |
| 34 | Vinorelbine             |  |  |  |  |  |
| 35 | All trans retinoic acid |  |  |  |  |  |
| 36 | Bortezomib              |  |  |  |  |  |
| 37 | Gefitinib               |  |  |  |  |  |
| 38 | Sorafenib               |  |  |  |  |  |
| 39 | Nilotinib               |  |  |  |  |  |
| 40 | Rituximab               |  |  |  |  |  |
| 41 | Trastuzumab             |  |  |  |  |  |
| 42 | Topotecan               |  |  |  |  |  |
| 43 | Filgastrim              |  |  |  |  |  |
| 44 | Lenalidomide            |  |  |  |  |  |
| 45 | Erlotinib               |  |  |  |  |  |
| 46 | Thalidomide             |  |  |  |  |  |
| 47 | Pertuzumab              |  |  |  |  |  |
| 48 | Cetuximab               |  |  |  |  |  |
| 49 | Bevacizumab             |  |  |  |  |  |
| 50 | Pembrolizumab           |  |  |  |  |  |

ii. By what mode do you procure your drugs?

Directly from distributors ☐

Open Tender ☐

Restricted Tender ☐

Importing ☐

Other (Specify). .....

iii. Are there different prices for different market segments?

Yes ☐ No ☐

iv. Have you witnessed a significant increase in the price of certain biologics due to an increase in their use?

Yes ☐ No ☐

If yes, please specify the drugs and comment on the price changes since you started stocking them.

| Name of Drug | Initial Price | New Price |
|--------------|---------------|-----------|
|              |               |           |
|              |               |           |
|              |               |           |

Research Assistant name:\_\_\_\_\_

Signature:\_\_\_\_\_

## 7.2 APPENDIX B. ETHICAL APPROVAL.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br><b>UNIVERSITY OF NAIROBI</b><br>FACULTY OF HEALTH SCIENCES<br>P O BOX 19676 Code 00202<br>Telegrams: varsity<br>Tel: (254-020) 2726308 Ext 44355                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>KNH-UoN ERC</b><br>Email: <a href="mailto:uonknh_erc@uonbi.ac.ke">uonknh_erc@uonbi.ac.ke</a><br>Website: <a href="http://www.erc.uonbi.ac.ke">http://www.erc.uonbi.ac.ke</a><br>Facebook: <a href="https://www.facebook.com/uonknh.erc">https://www.facebook.com/uonknh.erc</a><br>Twitter: @UONKNH_ERC <a href="https://twitter.com/UONKNH_ERC">https://twitter.com/UONKNH_ERC</a> | <br><b>KENYATTA NATIONAL HOSPITAL</b><br>P O BOX 20723 Code 00202<br>Tel: 726300-9<br>Fax: 725272<br>Telegrams: MEDSUP, Nairobi |
| Ref: KNH-ERC/A/487                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                        | 20 <sup>th</sup> December 2021                                                                                                                                                                                    |
| <p>Dr. Lewis Mutugi Mbaya<br/>Reg. No. U51/35847/2019<br/>Pharmacology and Pharmacognosy Unit<br/>Dept. of Pharmacy<br/>Faculty of Health Sciences<br/>University of Nairobi</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                   |
| <p>Dear Dr. Mbaya,</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                   |
| <p>RESEARCH PROPOSAL: AVAILABILITY, PRICE AND AFFORDABILITY OF ANTICANCER DRUGS IN NAIROBI, KENYA (P816/10/2021)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                   |
| <p>This is to inform you that KNH-UoN ERC has reviewed and approved your above research proposal. Your application approval number is <b>P816/10/2021</b>. The approval period is 20<sup>th</sup> December 2021 – 19<sup>th</sup> December 2022.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                   |
| <p>This approval is subject to compliance with the following requirements;</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                   |
| <ul style="list-style-type: none"><li>i. Only approved documents including (informed consents, study instruments, MTA) will be used</li><li>ii. All changes including (amendments, deviations, and violations) are submitted for review and approval by KNH-UoN ERC.</li><li>iii. Death and life threatening problems and serious adverse events or unexpected adverse events whether related or unrelated to the study must be reported to KNH-UoN ERC 72 hours of notification</li><li>iv. Any changes, anticipated or otherwise that may increase the risks or affected safety or welfare of study participants and others or affect the integrity of the research must be reported to KNH-UoN ERC within 72 hours</li><li>v. Clearance for export of biological specimens must be obtained from relevant institutions.</li><li>vi. Submission of a request for renewal of approval at least 60 days prior to expiry of the approval period. Attach a comprehensive progress report to support the renewal.</li><li>vii. Submission of an executive summary report within 90 days upon completion of the study to KNH-UoN ERC.</li></ul> |                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                   |
| <p>Protect to discover</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                   |



### 7.3 APPENDIX C: APPROVAL FOR WAIVER OF CONSENT.

#### 7.4 APPENDIX D: REQUEST FOR WAIVER OF CONSENT.



UNIVERSITY OF NAIROBI  
(UoN)  
COLLEGE OF HEALTH  
SCIENCES  
P O BOX 19676 Code 00202  
Telegrams: varsity  
(254-020) 2726300 Ext 44355

KNH-UoN ERC  
Email: [uonknh\\_erc@uonbi.ac.ke](mailto:uonknh_erc@uonbi.ac.ke)  
Website: <http://www.erc.uonbi.ac.ke>  
Facebook: <https://www.facebook.com/uonknh.erc>  
Twitter: @UONKNH\_ERC



KENYATTA NATIONAL  
HOSPITAL (KNH)  
P O BOX 20723 Code 00202  
Tel: 726300-9  
Fax: 725272  
Telegrams: MEDSUP, Nairobi

(To be submitted with Application for ERC Review of Research)  
Exempt studies to be defined

KNH-UoN ERC  
REQUEST FOR WAIVER OF INFORMED CONSENT  
(Not Required for Exempt Studies)



Project Title: AVAILABILITY, PRICE AND AFFORDABILITY OF ANTICANCER DRUGS  
IN NAIROBI, KENYA.

Principal Investigator and Institutional affiliation: MBAYA LEWIS MUTUGI, SCHOOL OF  
PHARMACY, UNIVERSITY OF NAIROBI

Date: 24<sup>th</sup> SEPTEMBER 2021

Under special circumstances, investigators may request one of three types of waivers to obtaining written informed consent from research participants.

1. Alteration of informed consent.

With this waiver, the investigator may provide to the participants a consent which does not include or which alters one or all of the required elements. Examples of when this waiver might be applicable would be, when a researcher is conducting secondary data analysis and the participants cannot be located or when requiring informed consent might somehow actually have negative consequences for research participants.

**2. Waiver of parental permission.**

This waiver would be used in cases where something may be legal for a child to do (i.e. contraception) without parental permission and obtaining parental permission would violate that privacy. An example of this type of waiver would be a survey on children (which would require parental permission) but the survey is about their experience on contraception usage.

**3. Waiver of written documentation that informed consent was obtained.** With this waiver, the investigator would be required to read or provide the informed consent form to a participant, but would not need to obtain the participant's signature on the consent form. Examples of when this waiver might be applicable would be some internet or phone surveys or when signing the form might have some negative consequence for the participant. It must be emphasized that these waivers will be given only when there are compelling reasons for doing so.

---

The Ethics and Research Committee determines which type of consent applies to your research, but please indicate the type that you are requesting.

☒ **Waiver or alteration of the informed consent process.** *(Complete Section I)*

☐ **Request for waiver of parental permission.** *(Complete Section II)*

☐ **Waiver of written documentation of consent.** *(Complete Section III)*



**I. Request for waiver or alteration of the consent process** (Not required for Exempt studies)

I believe that this protocol is eligible for waiver or alteration of required elements of the informed consent process because the protocol meets all of the following criteria: (Provide protocol-specific supporting information for each criterion that justifies the findings for the following :)

1. The research presents no more than “minimal risk” of harm to participants.

This study does not involve any intervention on the participants and seeks to obtain information on anticancer drug prices hence no harm will be dealt to the participants.

2. The waiver or alteration will not adversely affect the rights and welfare of the participants.

The right to privacy of the institutions will be upheld by not revealing their identities once the data has been collected. Instead, identifiers such as “Hospital 1” or “Supplier 1” shall be used to present the results.

3. The research could not practicably be carried out without the waiver or alteration.

The study seeks to establish prices of anticancer drugs within the market. Due to competitive practices, the different hospitals, suppliers and retailers may not be willing to share this information if made aware of the study. Waiver of consent will enable the researcher obtain this information objectively.

4. Whenever appropriate, the participants will be provided with additional pertinent information after participation.

While the researcher does not plan to disclose any information due to issues of bias, pertinent information will be disclosed where necessary on a case by case basis.

5. Elements of informed consent for which a waiver or alteration is requested and the rationale for each:

This research does not plan to inform participating institutions of the study as the researcher is seeking for the anticancer drug prices. The rationale for this is that, once informed, the institutions might withhold the required data or provide biased information that will then compromise the study.

6. The research does not involve non-viable neonates:



This research is a drug price study and does not involve non-viable neonates.

7. The research is not subject to FDA and/or national research regulation:

This research is a drug price study and is not subject to FDA and/or national research regulation.

**II. Request for waiver of parental permission** (Not required for Exempt studies)

I believe that this protocol is eligible for waiver of parental permission because the protocol meets all of the following criteria: (Provide protocol-specific supporting information for each criterion that justifies the findings for one of the following two options :)

**Option 1**

1. The research presents no more than "minimal risk" of harm to participants.

---

---

2. The waiver or alteration will not adversely affect the rights and welfare of the participants.

---

---

3. The research could not practicably be carried out without the waiver or alteration.

---

---



4. Whenever appropriate, the participants will be provided with additional pertinent information after participation.

\_\_\_\_\_

5. Elements of informed consent for which a waiver or alteration is requested and the rationale for each:

\_\_\_\_\_

6. The research does not involve non-viable neonates:

\_\_\_\_\_

7. The research is not subject to FDA and/or national research regulation:

\_\_\_\_\_

**Option 2:**

1. The research protocol is designed for conditions or for a participant population for which parental or guardian permission is not a reasonable requirement to protect the participants (for example, neglected or abused children)

\_\_\_\_\_

2. An appropriate mechanism for protecting the children who will participate as participant in the research will be substituted

\_\_\_\_\_

\_\_\_\_\_

3. The research is not subject to FDA and/or national research regulation:

\_\_\_\_\_

\_\_\_\_\_

4. The waiver is consistent with international and national law:

\_\_\_\_\_

**III. Request for waiver of written documentation of consent** (Not required for Exempt studies and not required when the consent process is waived.)

I believe that this protocol is eligible for a waiver of written documentation of informed consent because the protocol meets one of the following criteria: (Provide protocol-specific supporting information for each criterion that justifies the findings for one of the following two options :)  
**(NOTE: Even when documentation of informed consent is waived, the investigator is required to give participants full consent information, and to obtain their voluntary consent orally.)**

**Option 1**

*(Example: Conducting interviews with street children engaged in drug abuse. The only record of the name or other identifying information of the participants would be the signed consent form and knowledge of an individual's participation or information provided could lead to potential legal, social, or physical harm.)*

Explain:



1. The only record linking the participant and the research would be the consent document.

---

---

---

2. The principle risk would be potential harm resulting from breach of confidentiality.

---

---

---

3. Each participant will be asked whether the subject wants documentation linking the participant with the research and the participant's wishes will govern.

---

---

---

4. The research is not subject to FDA and / national research regulation.

---

---

---

#### Option 2

*(Example: Using an anonymous survey consent or conducting telephone interviews with politicians about how constitutional provision for funding of political parties will affecting the campaign process of smaller parties*

1. The research presents no more than minimal risk of harm to participants.

---

---

---

2. The research involves no procedures for which written consent is normally required outside of the research context.

Approval (KNH-UoN ERC Chairperson: Check all that apply to indicate that the waiver or alteration is approved and to indicate agreement with the investigators protocol specific findings justifying the waiver.)

☐ Waiver or Alteration of the Consent Process

☐ Waiver of parental permission

☐ Waiver of Written Documentation of Consent

**NOTE:** To approve a waiver of written documentation of informed consent the investigator must provide a written document describing the information to be disclosed. This document has to include all required and appropriate additional elements of consent disclosure, unless the consent process has been altered.

Chose one of the following when approving a waiver of written documentation:

☐ The investigator must provide a written description of the information provided orally to the participant.

☐ The investigator does not have to provide a written description of the information provided orally to the participant.

APPROVED BY CHAIR KNH-UoN ERC:

Name: Amay  
8/6/2023

Signature Angela Amay



## 7.4 APPENDIX D: AVAILABILITY OF ANTICANCER DRUGS.

Table 7.1 Availability of Molecular Targeted Agents, Enzymes and Other Antineoplastic Agents

| Medicine Name               | Strength | Percent Availability % |                 |     |          |                   |             |
|-----------------------------|----------|------------------------|-----------------|-----|----------|-------------------|-------------|
|                             |          | Tier 1 Supplier        | Tier 2 Supplier | KNH | Agencies | Private Hospitals | Average     |
| Rituximab                   | 500mg    | 100                    | 62.5            | 100 | 50       | 66.7              | 73.7        |
| Trastuzumab                 | 440mg    | 100                    | 50              | 100 | 50       | 100.0             | 73.7        |
| Bevacizumab                 | 100mg    | 80                     | 50              | 100 | 50       | 100.0             | 63.2        |
| Bortezomib                  | 3.5mg    | 40                     | 62.5            | 100 | 50       | 100.0             | 57.9        |
| Rituximab                   | 100mg    | 80                     | 50              | 0   | 0        | 66.7              | 57.9        |
| Bevacizumab                 | 400mg    | 80                     | 50              | 0   | 0        | 100.0             | 57.9        |
| Filgastrim                  | 300mcg   | 40                     | 50              | 100 | 50       | 66.7              | 52.6        |
| Asparaginase                | 10.000IU | 20                     | 62.5            | 100 | 50       | 33.3              | 47.4        |
| Gefitinib                   | 250mg    | 0                      | 62.5            | 100 | 50       | 100.0             | 47.4        |
| Lenalidomide                | 25mg     | 40                     | 50              | 0   | 0        | 100.0             | 47.4        |
| Lenalidomide                | 10mg     | 40                     | 37.5            | 100 | 50       | 66.7              | 42.1        |
| Thalidomide                 | 100mg    | 40                     | 25              | 100 | 100      | 66.7              | 42.1        |
| Cetuximab                   | 100mg    | 20                     | 37.5            | 100 | 50       | 66.7              | 36.8        |
| Erlotinib                   | 150mg    | 20                     | 62.5            | 0   | 0        | 33.3              | 36.8        |
| Bortezomib                  | 2mg      | 20                     | 50              | 0   | 0        | 33.3              | 31.6        |
| imatinib                    | 400mg    | 40                     | 37.5            | 0   | 0        | 33.3              | 31.6        |
| Trastuzumab                 | 150mg    | 60                     | 25              | 0   | 0        | 33.3              | 31.6        |
| pembrolizumab               | 100mg    | 20                     | 37.5            | 0   | 0        | 66.7              | 31.6        |
| Sorafenib                   | 200mg    | 20                     | 37.5            | 0   | 0        | 66.7              | 31.6        |
| Nilotinib                   | 200mg    | 40                     | 12.5            | 100 | 50       | 33.3              | 26.3        |
| Trastuzumab                 | 600mg    | 60                     | 0               | 0   | 0        | 66.7              | 26.3        |
| imatinib                    | 100mg    | 20                     | 37.5            | 0   | 0        | 0.0               | 21.1        |
| Nilotinib                   | 150mg    | 40                     | 12.5            | 0   | 0        | 0.0               | 15.8        |
| Pertuzumab                  | 420mg    | 0                      | 25              | 0   | 0        | 33.3              | 15.8        |
| Cetuximab                   | 500mg    | 20                     | 12.5            | 0   | 0        | 0.0               | 10.5        |
| Thalidomide                 | 50mg     | 20                     | 0               | 0   | 0        | 0.0               | 5.3         |
| Arsenic Trioxide            | 1mg      | 0                      | 0               | 0   | 0        | 0.0               | 0.0         |
| <b>Average Availability</b> |          |                        |                 |     |          |                   | <b>37.6</b> |

Table 7.2 Availability of Alkylating Agents, Antimetabolites and Detoxifying Agents

| Medicine Name      | Strength        | Percent Availability (%) |                 |     |          |                   |         |
|--------------------|-----------------|--------------------------|-----------------|-----|----------|-------------------|---------|
|                    |                 | Tier 1 Supplier          | Tier 2 Supplier | KNH | Agencies | Private Hospitals | Average |
| Carboplatin        | 450mg           | 60                       | 87.5            | 100 | 50       | 100.0             | 78.9    |
| Capecitabine       | 500mg           | 100                      | 50              | 100 | 50       | 100.0             | 73.7    |
| Cisplatin          | 50mg            | 40                       | 87.5            | 100 | 50       | 100.0             | 73.7    |
| Gemcitabine        | 1000mg          | 40                       | 87.5            | 0   | 100      | 100.0             | 73.7    |
| Oxaliplatin        | 100mg           | 60                       | 75              | 100 | 50       | 100.0             | 73.7    |
| Dacarbazine        | 200mg           | 40                       | 75              | 100 | 50       | 66.7              | 63.2    |
| Carboplatin        | 150mg           | 60                       | 75              | 0   | 50       | 33.3              | 57.9    |
| Fluorouracil       | 500mg           | 40                       | 75              | 0   | 0        | 100.0             | 57.9    |
| Methotrexate       | 50mg            | 60                       | 50              | 0   | 50       | 100.0             | 57.9    |
| Oxaliplatin        | 50mg            | 60                       | 62.5            | 0   | 50       | 66.7              | 57.9    |
| Cyclophosphamide   | 500mg           | 40                       | 62.5            | 100 | 50       | 33.3              | 52.6    |
| Cyclophosphamide   | 1000mg          | 20                       | 75              | 0   | 50       | 66.7              | 52.6    |
| Gemcitabine        | 200mg           | 40                       | 50              | 100 | 100      | 33.3              | 52.6    |
| Chlorambucil       | 2mg             | 40                       | 37.5            | 100 | 50       | 66.7              | 47.4    |
| Cytarabine         | 100mg           | 20                       | 37.5            | 100 | 100      | 66.7              | 47.4    |
| Melphalan          | 2mg             | 40                       | 37.5            | 100 | 50       | 66.7              | 47.4    |
| Mercaptopurine     | 50mg            | 40                       | 25              | 100 | 100      | 66.7              | 47.4    |
| Calcium folinate   | 50mg            | 20                       | 37.5            | 0   | 50       | 100.0             | 42.1    |
| Ifosfamide + mesna | 1000mg + 600mg  | 0                        | 50              | 100 | 50       | 66.7              | 42.1    |
| Bendamustine       | 100mg           | 40                       | 25              | 100 | 0        | 66.7              | 36.8    |
| Capecitabine       | 150mg           | 60                       | 0               | 0   | 50       | 100.0             | 36.8    |
| Cyclophosphamide   | 200mg           | 0                        | 50              | 0   | 100      | 33.3              | 36.8    |
| Fluorouracil       | 250mg           | 20                       | 50              | 100 | 50       | 0.0               | 36.8    |
| Cytarabine         | 500mg           | 40                       | 25              | 0   | 50       | 33.3              | 31.6    |
| Cytarabine         | 1000mg          | 40                       | 37.5            | 0   | 0        | 33.3              | 31.6    |
| Methotrexate       | 2.5mg           | 20                       | 25              | 100 | 50       | 33.3              | 31.6    |
| Procarbazine       | 50mg            | 40                       | 0               | 100 | 50       | 66.7              | 31.6    |
| Cisplatin          | 10mg            | 20                       | 50              | 0   | 0        | 0.0               | 26.3    |
| Hydroxycarbamide   | 500mg           | 0                        | 25              | 100 | 50       | 33.3              | 26.3    |
| Ifosfamide + mesna | 2000mg + 1200mg | 0                        | 37.5            | 0   | 50       | 33.3              | 26.3    |
| Dacarbazine        | 500mg           | 0                        | 37.5            | 0   | 0        | 33.3              | 21.1    |
| Calcium folinate   | 15mg            | 20                       | 0               | 100 | 0        | 33.3              | 15.8    |
| Bendamustine       | 25mg            | 0                        | 25              | 0   | 0        | 0.0               | 10.5    |
| Chlorambucil       | 5mg             | 20                       | 12.5            | 0   | 0        | 0.0               | 10.5    |

|                             |       |   |      |   |   |     |             |
|-----------------------------|-------|---|------|---|---|-----|-------------|
| Cisplatin                   | 100mg | 0 | 12.5 | 0 | 0 | 0.0 | 5.3         |
| Melphalan                   | 5mg   | 0 | 12.5 | 0 | 0 | 0.0 | 5.3         |
| Melphalan                   | 50mg  | 0 | 12.5 | 0 | 0 | 0.0 | 5.3         |
| Cyclophosphamide            | 50mg  | 0 | 0    | 0 | 0 | 0.0 | 0.0         |
| Tioguanine                  | 40mg  | 0 | 0    | 0 | 0 | 0.0 | 0.0         |
| <b>Average Availability</b> |       |   |      |   |   |     | <b>39.6</b> |

Table 7.3 Availability of Plant Derived Agents, Antitumor Antibiotics and Retinoids

| Medicine Name           | Strength | Percentage Availability (%) |                 |     |          |                   |         |
|-------------------------|----------|-----------------------------|-----------------|-----|----------|-------------------|---------|
|                         |          | Tier 1 Supplier             | Tier 2 Supplier | KNH | Agencies | Private Hospitals | Average |
| Docetaxel               | 80mg     | 60                          | 75              | 100 | 50       | 100.0             | 73.7    |
| Doxorubicin             | 50mg     | 60                          | 62.5            | 100 | 100      | 100.0             | 73.7    |
| Dactinomycin            | 0.5mg    | 40                          | 25              | 200 | 200      | 66.7              | 63.2    |
| Etoposide               | 100mg    | 20                          | 75              | 100 | 50       | 100.0             | 63.2    |
| Paclitaxel              | 100mg    | 60                          | 75              | 100 | 50       | 33.3              | 63.2    |
| Vinblastine             | 10mg     | 20                          | 75              | 100 | 100      | 66.7              | 63.2    |
| Vincristine             | 1mg      | 60                          | 50              | 100 | 50       | 100.0             | 63.2    |
| Bleomycin               | 15iu     | 40                          | 62.5            | 100 | 50       | 66.7              | 57.9    |
| Irinotecan              | 100mg    | 40                          | 62.5            | 100 | 0        | 100.0             | 57.9    |
| Paclitaxel              | 300mg    | 40                          | 75              | 0   | 50       | 66.7              | 57.9    |
| Liposomal Doxorubicin   | 20mg     | 80                          | 37.5            | 100 | 0        | 66.7              | 52.6    |
| Daunorubicin            | 20mg     | 40                          | 37.5            | 100 | 100      | 33.3              | 47.4    |
| Docetaxel               | 20mg     | 40                          | 62.5            | 0   | 50       | 0.0               | 42.1    |
| Doxorubicin             | 10mg     | 40                          | 62.5            | 0   | 0        | 33.3              | 42.1    |
| Paclitaxel              | 30mg     | 20                          | 62.5            | 0   | 50       | 33.3              | 42.1    |
| Irinotecan              | 40mg     | 60                          | 37.5            | 0   | 50       | 0.0               | 36.8    |
| Vinorelbine             | 50mg     | 20                          | 12.5            | 100 | 50       | 100.0             | 36.8    |
| Docetaxel               | 120mg    | 20                          | 37.5            | 0   | 0        | 66.7              | 31.6    |
| Etoposide               | 50mg     | 40                          | 12.5            | 0   | 0        | 100.0             | 31.6    |
| Liposomal Doxorubicin   | 50mg     | 20                          | 25              | 0   | 0        | 33.3              | 21.1    |
| Paclitaxel              | 260mg    | 40                          | 25              | 0   | 0        | 0.0               | 21.1    |
| Vinorelbine             | 10mg     | 20                          | 25              | 0   | 0        | 0.0               | 15.8    |
| Topotecan               | 2.5mg    | 0                           | 25              | 100 | 0        | 0.0               | 15.8    |
| Daunorubicin            | 10mg     | 0                           | 0               | 0   | 0        | 33.3              | 5.3     |
| Vincristine             | 2mg      | 0                           | 12.5            | 0   | 0        | 0.0               | 5.3     |
| All trans retinoic acid | 10mg     | 0                           | 0               | 0   | 0        | 33.3              | 5.3     |

|                             |  |  |  |  |  |  |             |
|-----------------------------|--|--|--|--|--|--|-------------|
| <b>Average Availability</b> |  |  |  |  |  |  | <b>41.9</b> |
|-----------------------------|--|--|--|--|--|--|-------------|

Table 7.4 Availability of Originator Brands and Lowest Priced Generics

|                      |                 | <b>Hospitals (n=4)</b> |            | <b>Suppliers (n = 13)</b> |            | <b>Agencies (n=2)</b> |            |
|----------------------|-----------------|------------------------|------------|---------------------------|------------|-----------------------|------------|
| <b>Medicine Name</b> | <b>Strength</b> | <b>OB</b>              | <b>LPG</b> | <b>OB</b>                 | <b>LPG</b> | <b>OB</b>             | <b>LPG</b> |
| Arsenic trioxide     | 1mg             | 0 (0)                  | 0 (0)      | 0 (0)                     | 0 (0)      | 0 (0)                 | 0 (0)      |
| Asparaginase         | 10,000 IU       | 0 (0)                  | 2 (50)     | 0 (0)                     | 6 (46.15)  | 0 (0)                 | 1 (50)     |
| Bendamustine         | 25mg            | 0 (0)                  | 0 (0)      | 0 (0)                     | 2 (15.38)  | 0 (0)                 | 0 (0)      |
| Bendamustine         | 100mg           | 0 (0)                  | 3 (75)     | 0 (0)                     | 4 (30.77)  | 0 (0)                 | 0 (0)      |
| Bleomycin            | 15iu            | 0 (0)                  | 3 (75)     | 0 (0)                     | 7 (53.85)  | 0 (0)                 | 1 (50)     |
| Calcium folinate     | 15mg            | 0 (0)                  | 2 (50)     | 0 (0)                     | 1 (7.69)   | 0 (0)                 | 0 (0)      |
| Calcium folinate     | 50mg            | 0 (0)                  | 3 (75)     | 0 (0)                     | 4 (30.77)  | 0 (0)                 | 1 (50)     |
| Capecitabine         | 150mg           | 2 (50)                 | 1 (25)     | 3 (23.08)                 | 0 (0)      | 0 (0)                 | 1 (50)     |
| Capecitabine         | 500mg           | 2 (50)                 | 2 (50)     | 3 (23.08)                 | 6 (46.15)  | 0 (0)                 | 1 (50)     |
| Carboplatin          | 150mg           | 0 (0)                  | 1 (25)     | 0 (0)                     | 8 (61.54)  | 0 (0)                 | 1 (50)     |
| Carboplatin          | 450mg           | 0 (0)                  | 4 (100)    | 0 (0)                     | 9 (69.23)  | 0 (0)                 | 1 (50)     |
| Chlorambucil         | 2mg             | 2 (50)                 | 1 (25)     | 0 (0)                     | 5 (38.46)  | 0 (0)                 | 1 (50)     |
| Chlorambucil         | 5mg             | 0 (0)                  | 0 (0)      | 0 (0)                     | 2 (15.38)  | 0 (0)                 | 0 (0)      |
| Cisplatin            | 10mg            | 0 (0)                  | 0 (0)      | 0 (0)                     | 4 (30.77)  | 0 (0)                 | 0 (0)      |
| Cisplatin            | 50mg            | 0 (0)                  | 4 (100)    | 0 (0)                     | 8 (61.54)  | 0 (0)                 | 1 (50)     |
| Cisplatin            | 100mg           | 0 (0)                  | 0 (0)      | 0 (0)                     | 1 (7.69)   | 0 (0)                 | 0 (0)      |
| Cyclophosphamide     | 50mg            | 0 (0)                  | 2 (50)     | 0 (0)                     | 0 (0)      | 0 (0)                 | 0 (0)      |
| Cyclophosphamide     | 200mg           | 0 (0)                  | 2 (50)     | 0 (0)                     | 4 (30.77)  | 0 (0)                 | 2 (100)    |
| Cyclophosphamide     | 500mg           | 0 (0)                  | 2 (50)     | 0 (0)                     | 7 (53.85)  | 0 (0)                 | 1 (50)     |
| Cyclophosphamide     | 1000mg          | 0 (0)                  | 3 (75)     | 0 (0)                     | 7 (53.85)  | 0 (0)                 | 1 (50)     |
| Cytarabine           | 100mg           | 1 (25)                 | 0 (0)      | 1 (7.69)                  | 3 (23.08)  | 0 (0)                 | 2 (100)    |
| Cytarabine           | 500mg           | 0 (0)                  | 0 (0)      | 0 (0)                     | 4 (30.77)  | 0 (0)                 | 1 (50)     |
| Cytarabine           | 1000mg          | 0 (0)                  | 3 (75)     | 0 (0)                     | 5 (38.46)  | 0 (0)                 | 0 (0)      |
| Dacarbazine          | 200mg           | 0 (0)                  | 3 (75)     | 0 (0)                     | 8 (61.54)  | 0 (0)                 | 1 (50)     |
| Dacarbazine          | 500mg           | 0 (0)                  | 1 (25)     | 0 (0)                     | 3 (23.08)  | 0 (0)                 | 0 (0)      |
| Dactinomycin         | 0.5mg           | 0 (0)                  | 3 (75)     | 0 (0)                     | 4 (30.77)  | 0 (0)                 | 1 (50)     |
| Daunorubicin         | 10mg            | 0 (0)                  | 1 (25)     | 0 (0)                     | 0 (0)      | 0 (0)                 | 0 (0)      |
| Daunorubicin         | 20mg            | 0 (0)                  | 2 (50)     | 0 (0)                     | 5 (38.46)  | 0 (0)                 | 2 (100)    |
| Docetaxel            | 20mg            | 0 (0)                  | 0 (0)      | 1 (7.69)                  | 7 (53.85)  | 0 (0)                 | 0 (0)      |

|                          |                    |        |         |           |           |       |         |
|--------------------------|--------------------|--------|---------|-----------|-----------|-------|---------|
| Docetaxel                | 80mg               | 1 (25) | 3 (75)  | 1 (7.69)  | 9 (69.23) | 0 (0) | 1 (50)  |
| Docetaxel                | 120mg              | 0 (0)  | 2 (50)  | 0 (0)     | 3 (23.08) | 0 (0) | 0 (0)   |
| Doxorubicin              | 10mg               | 0 (0)  | 1 (25)  | 0 (0)     | 7 (53.85) | 0 (0) | 0 (0)   |
| Doxorubicin              | 50mg               | 0 (0)  | 4 (100) | 1 (7.69)  | 7 (53.85) | 0 (0) | 2 (100) |
| Etoposide                | 50mg               | 0 (0)  | 3 (75)  | 0 (0)     | 3 (23.08) | 0 (0) | 0 (0)   |
| Etoposide                | 100mg              | 0 (0)  | 4 (100) | 0 (0)     | 7 (53.85) | 0 (0) | 1 (50)  |
| Fluorouracil             | 250mg              | 0 (0)  | 1 (25)  | 0 (0)     | 5 (38.46) | 0 (0) | 1 (50)  |
| Fluorouracil             | 500mg              | 0 (0)  | 3 (75)  | 0 (0)     | 8 (61.54) | 0 (0) | 0 (0)   |
| Gemcitabine              | 200mg              | 0 (0)  | 1 (25)  | 0 (0)     | 6 (46.15) | 0 (0) | 2 (100) |
| Gemcitabine              | 1000mg             | 0 (0)  | 3 (75)  | 0 (0)     | 9 (69.23) | 0 (0) | 2 (100) |
| Hydroxycarbamide         | 500mg              | 1 (25) | 1 (25)  | 0 (0)     | 1 (7.69)  | 0 (0) | 1 (50)  |
| Ifosfamide + mesna       | 1000mg +<br>600mg  | 0 (0)  | 3 (75)  | 0 (0)     | 4 (30.77) | 0 (0) | 1 (50)  |
| Ifosfamide + mesna       | 2000mg +<br>1200mg | 0 (0)  | 1 (25)  | 0 (0)     | 3 (23.08) | 0 (0) | 1 (50)  |
| Irinotecan               | 40mg               | 0 (0)  | 0 (0)   | 0 (0)     | 4 (30.77) | 0 (0) | 1 (50)  |
| Irinotecan               | 100mg              | 0 (0)  | 4 (100) | 1 (7.69)  | 8 (61.54) | 0 (0) | 0 (0)   |
| Liposomal<br>Doxorubicin | 20mg               | 0 (0)  | 3 (75)  | 2 (15.38) | 5 (38.46) | 0 (0) | 0 (0)   |
| Liposomal<br>Doxorubicin | 50mg               | 0 (0)  | 2 (50)  | 0 (0)     | 3 (23.08) | 0 (0) | 0 (0)   |
| Melphalan                | 2mg                | 0 (0)  | 2 (50)  | 0 (0)     | 5 (38.46) | 0 (0) | 1 (50)  |
| Melphalan                | 5mg                | 0 (0)  | 0 (0)   | 0 (0)     | 1 (7.69)  | 0 (0) | 0 (0)   |
| Melphalan                | 50mg               | 0 (0)  | 0 (0)   | 0 (0)     | 1 (7.69)  | 0 (0) | 0 (0)   |
| Mercaptopurine           | 50mg               | 1 (25) | 2 (50)  | 0 (0)     | 4 (30.77) | 0 (0) | 2 (100) |
| Methotrexate             | 2.5mg              | 0 (0)  | 2 (50)  | 0 (0)     | 3 (23.08) | 0 (0) | 1 (50)  |
| Methotrexate             | 50mg               | 0 (0)  | 3 (75)  | 0 (0)     | 7 (53.85) | 0 (0) | 1 (50)  |
| Oxaliplatin              | 50mg               | 1 (25) | 2 (50)  | 1 (7.69)  | 8 (61.54) | 0 (0) | 1 (50)  |
| Oxaliplatin              | 100mg              | 1 (25) | 4 (100) | 1 (7.69)  | 9 (69.23) | 0 (0) | 1 (50)  |
| Paclitaxel               | 30mg               | 0 (0)  | 1 (25)  | 0 (0)     | 6 (46.15) | 0 (0) | 1 (50)  |
| Paclitaxel               | 100mg              | 0 (0)  | 3 (75)  | 0 (0)     | 9 (69.23) | 0 (0) | 1 (50)  |
| Paclitaxel               | 260mg              | 0 (0)  | 2 (50)  | 0 (0)     | 4 (30.77) | 0 (0) | 0 (0)   |
| Paclitaxel               | 300mg              | 0 (0)  | 2 (50)  | 0 (0)     | 8 (61.54) | 0 (0) | 1 (50)  |
| Procarbazine             | 50mg               | 0 (0)  | 3 (75)  | 0 (0)     | 2 (15.38) | 0 (0) | 1 (50)  |
| Tioguanine               | 40mg               | 0 (0)  | 0 (0)   | 0 (0)     | 0 (0)     | 0 (0) | 0 (0)   |
| Vinblastine              | 10mg               | 0 (0)  | 3 (75)  | 0 (0)     | 7 (53.85) | 0 (0) | 2 (100) |
| Vincristine              | 1mg                | 0 (0)  | 4 (100) | 0 (0)     | 7 (53.85) | 0 (0) | 1 (50)  |
| Vincristine              | 2mg                | 0 (0)  | 0 (0)   | 0 (0)     | 1 (7.69)  | 0 (0) | 0 (0)   |
| Vinorelbine              | 10mg               | 0 (0)  | 0 (0)   | 0 (0)     | 4 (30.77) | 0 (0) | 0 (0)   |
| Vinorelbine              | 50mg               | 0 (0)  | 4 (100) | 0 (0)     | 2 (15.38) | 0 (0) | 1 (50)  |
| All trans retinoic acid  | 10mg               | 0 (0)  | 1 (25)  | 0 (0)     | 0 (0)     | 0 (0) | 0 (0)   |
| Bortezomib               | 2mg                | 0 (0)  | 1 (25)  | 0 (0)     | 5 (38.46) | 0 (0) | 0 (0)   |
| Bortezomib               | 3.5mg              | 0 (0)  | 4 (100) | 1 (7.69)  | 6 (46.15) | 0 (0) | 1 (50)  |

|                             |        |              |               |              |               |              |            |
|-----------------------------|--------|--------------|---------------|--------------|---------------|--------------|------------|
| Gefitinib                   | 250mg  | 0 (0)        | 4 (100)       | 0 (0)        | 5 (38.46)     | 0 (0)        | 0 (0)      |
| Imatinib                    | 100mg  | 0 (0)        | 0 (0)         | 0 (0)        | 4 (30.77)     | 0 (0)        | 0 (0)      |
| Imatinib                    | 400mg  | 0 (0)        | 1 (25)        | 1 (7.69)     | 5 (38.46)     | 0 (0)        | 0 (0)      |
| Nilotinib                   | 150mg  | 1 (25)       | 0 (0)         | 0 (0)        | 2 (15.38)     | 0 (0)        | 0 (0)      |
| Nilotinib                   | 200mg  | 0 (0)        | 1 (25)        | 2 (15.38)    | 2 (15.38)     | 0 (0)        | 0 (0)      |
| Rituximab                   | 100mg  | 1 (25)       | 2 (50)        | 1 (7.69)     | 8 (61.54)     | 0 (0)        | 1 (50)     |
| Rituximab                   | 500mg  | 2 (50)       | 2 (50)        | 2 (15.38)    | 9 (69.23)     | 0 (0)        | 1 (50)     |
| Trastuzumab                 | 150mg  | 1 (25)       | 1 (25)        | 3 (23.08)    | 2 (15.38)     | 0 (0)        | 0 (0)      |
| Trastuzumab                 | 440mg  | 2 (50)       | 3 (75)        | 3 (23.08)    | 6 (46.15)     | 1 (50)       | 1 (50)     |
| Trastuzumab                 | 600mg  | 2 (50)       | 0 (0)         | 3 (23.08)    | 0 (0)         | 0 (0)        | 0 (0)      |
| Topotecan                   | 2.5mg  | 0 (0)        | 1 (25)        | 0 (0)        | 1 (7.69)      | 0 (0)        | 0 (0)      |
| Filgastrim                  | 300mcg | 0 (0)        | 3 (75)        | 0 (0)        | 6 (46.15)     | 0 (0)        | 1 (50)     |
| Lenalidomide                | 10mg   | 0 (0)        | 4 (100)       | 0 (0)        | 5 (38.46)     | 0 (0)        | 0 (0)      |
| Lenalidomide                | 25mg   | 0 (0)        | 3 (75)        | 0 (0)        | 6 (46.15)     | 0 (0)        | 0 (0)      |
| Pembrolizumab               | 100mg  | 0 (0)        | 2 (50)        | 3 (23.08)    | 2 (15.38)     | 0 (0)        | 0 (0)      |
| Thalidomide                 | 100mg  | 0 (0)        | 3 (75)        | 0 (0)        | 1 (7.69)      | 0 (0)        | 0 (0)      |
| Thalidomide                 | 50mg   | 0 (0)        | 0 (0)         | 0 (0)        | 4 (30.77)     | 0 (0)        | 1 (50)     |
| Pertuzumab                  | 420mg  | 1 (25)       | 0 (0)         | 2 (15.38)    | 0 (0)         | 0 (0)        | 0 (0)      |
| Cetuximab                   | 100mg  | 0 (0)        | 2 (50)        | 1 (7.69)     | 3 (23.08)     | 0 (0)        | 0 (0)      |
| Cetuximab                   | 500mg  | 0 (0)        | 0 (0)         | 1 (7.69)     | 1 (7.69)      | 0 (0)        | 0 (0)      |
| Bevacizumab                 | 100mg  | 1 (25)       | 3 (75)        | 3 (23.08)    | 6 (46.15)     | 0 (0)        | 0 (0)      |
| Bevacizumab                 | 400mg  | 1 (25)       | 4 (100)       | 3 (23.08)    | 6 (46.15)     | 0 (0)        | 0 (0)      |
| <b>Average Availability</b> |        | <b>6.70%</b> | <b>46.90%</b> | <b>3.70%</b> | <b>34.10%</b> | <b>0.50%</b> | <b>30%</b> |

Value in bracket denotes percentage availability for each drug per provider.